

Compal confidential

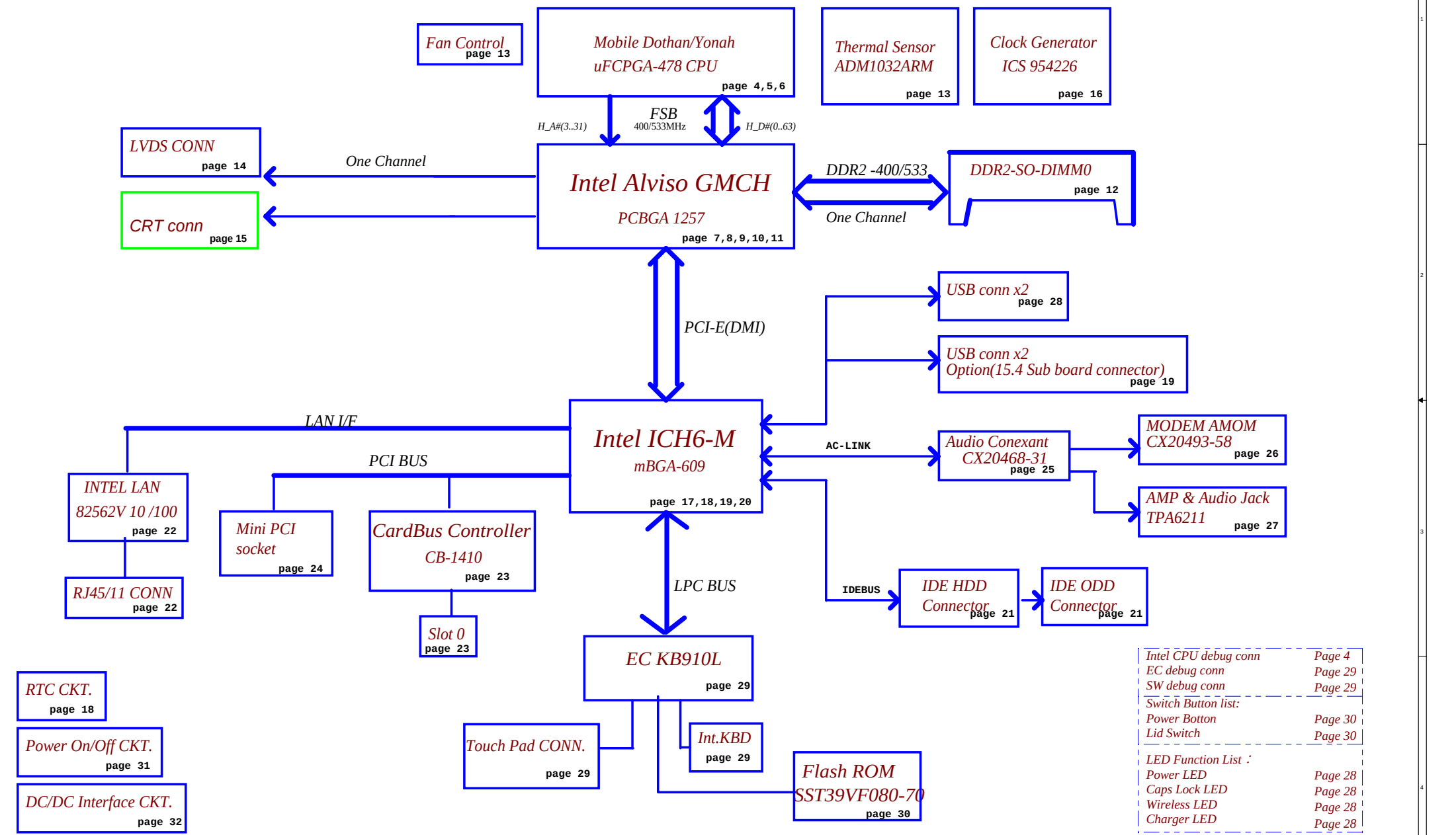
Schematics Document Mobile Dothan uFCPGA with Intel Alviso_GM+ICH6-M core logic

2006-09-15

REV:1.0

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I2C / SMBUS ADDRESSING

External PCI Devices

DEVICE	IDSEL #	REQ/GNT #	PIRQ
CARD BUS	AD22	2	C
Wireless LAN(MINI PCI)	AD20	0	E, F

Power Managment table

Signal State	+3VALW +5VALW	+1.8V	+CPU CORE +VCCP(1.05V) +5VS +3VS +2.5VS +1.5VS +1.8VS +0.9VS
S0	ON	ON	ON
S1	ON	ON	ON
S3	ON	ON	OFF
S5 S4/AC	ON	OFF	OFF
S5 S4/AC don't exist	OFF	OFF	OFF

SMBUS Control Table

	SOURCE	INVERTER	BATT	SERIAL EEPROM	THERMAL SENSOR (CPU) ADM1032	SODIMM	CLK CHIP	MINI PCI	LCD
SMB_EC_CK1 SMB_EC_DA1	KB910L	×	✓	×	×	×	×	×	×
SMB_EC_CK2 SMB_EC_DA2	KB910L	×	×	×	✓	×	×	×	×
ICH_SMBCLK ICH_SMBDATA	ICH6-M	×	×	×	×	✓	✓	×	×
LCD_DDCCLK LCD_DDCDATA	Alviso GM-GP	×	×	×	×	×	×	×	✓
I2CC_SCL I2CC_SDA	NV44M	×	×	×	×	×	×	×	✓

Alviso 915GM SA00000K040

Alviso 910GML SA00000K100

BOM

@ : not install

45@ : 45 level

14@ : 14"(IAT00) install

WLAN@ : with WLAN (mini PCI) install

AMOM@ : with AMOM install

BATT@ : 45 level

SPI@ : SPI ROM install

conn@ : ME part

IAT00 14"

46144232L01 915GM

46144232L02 910GML W/O WLAN

46144232L03 910GML

IAT10 15.4"

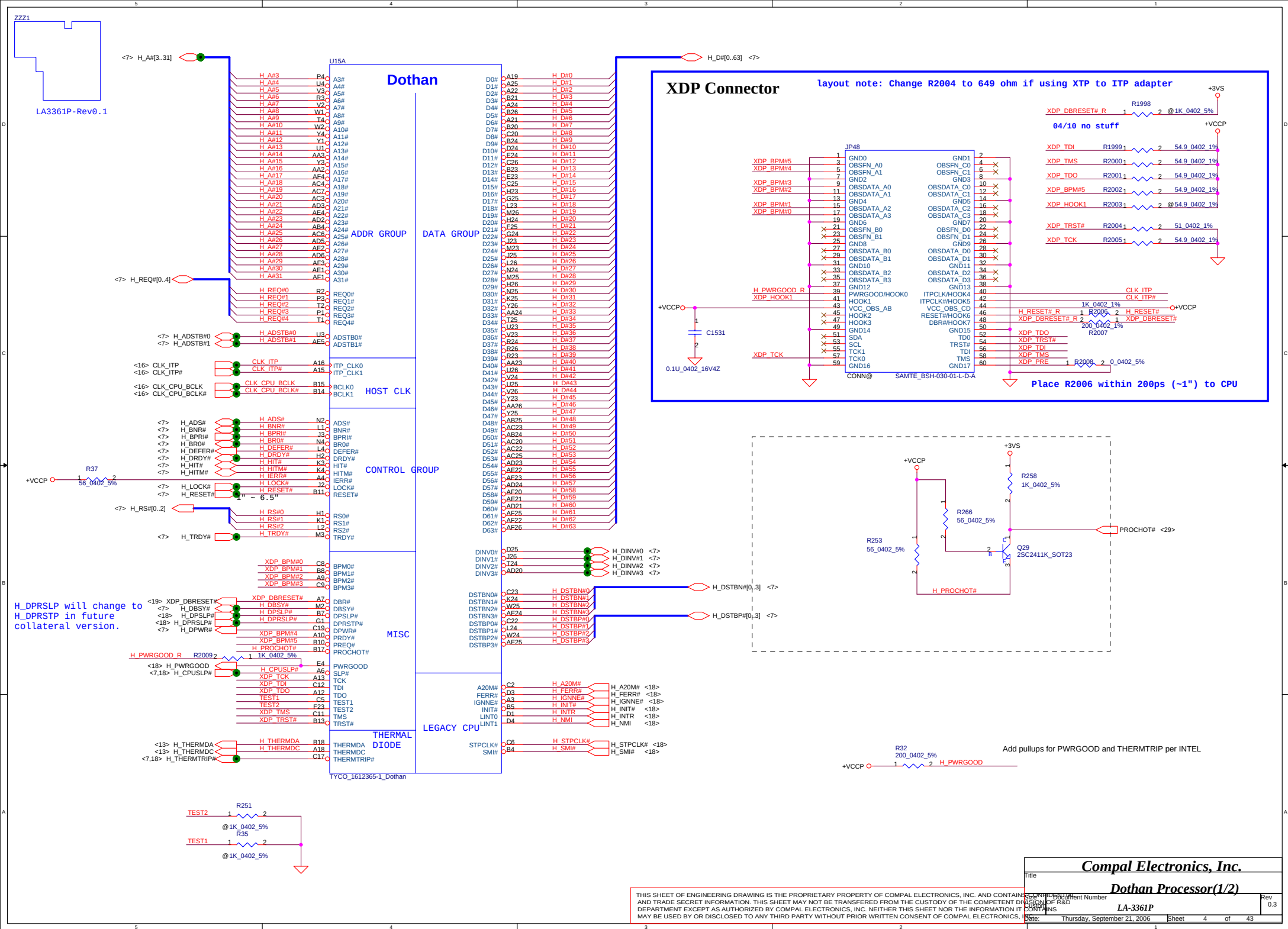
46144232L11 910GML

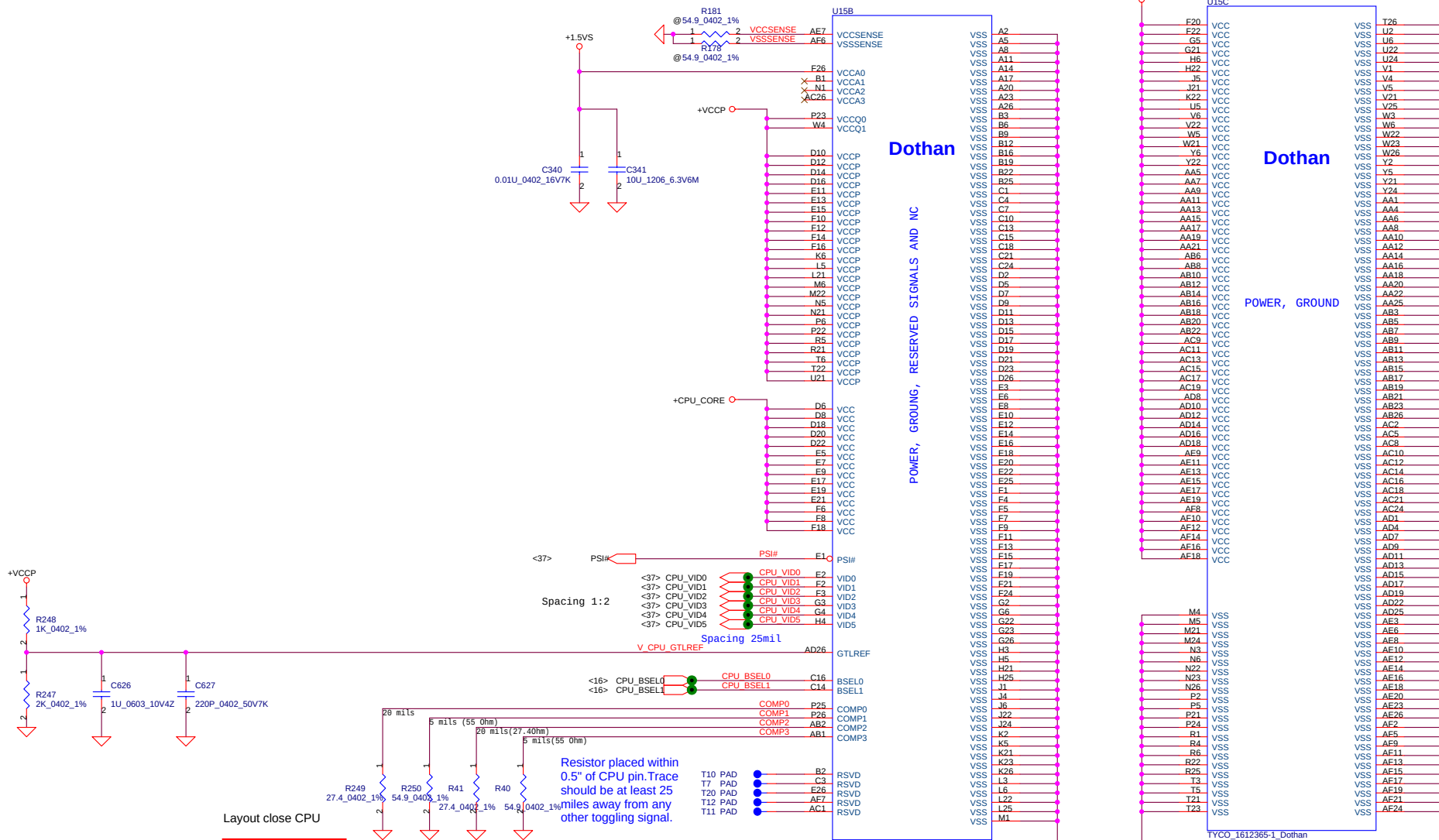
46144232L12 915GM

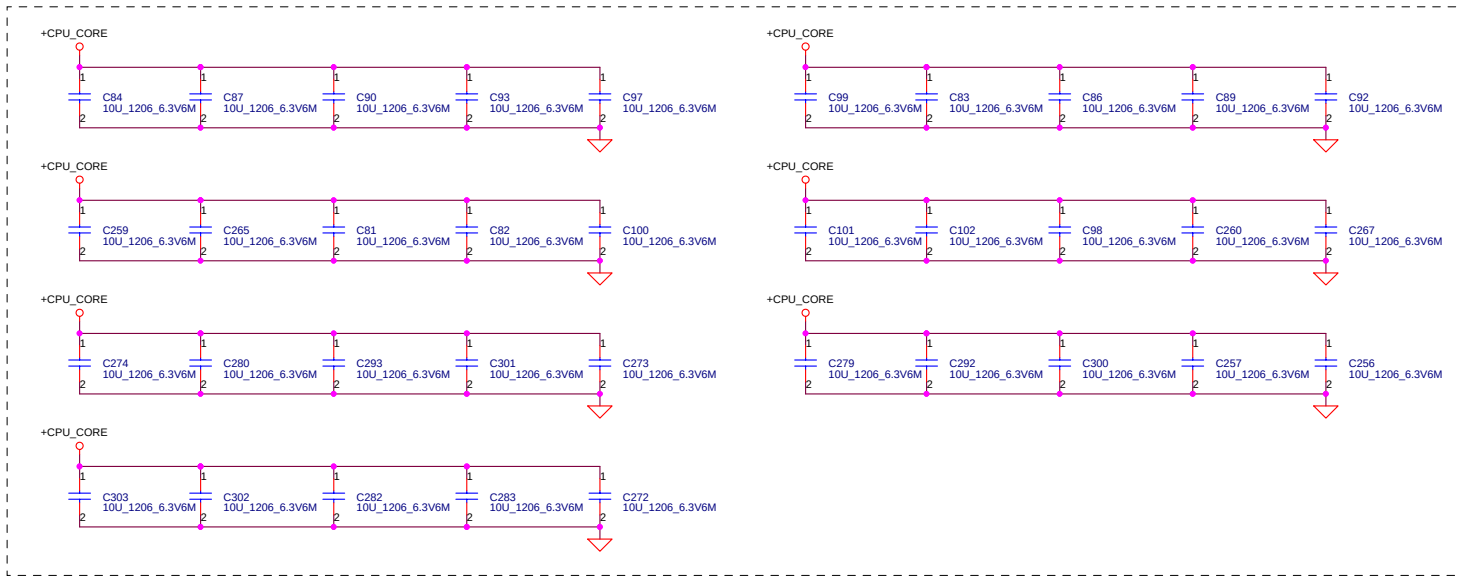
46144232L13 910GML W/O WLAN

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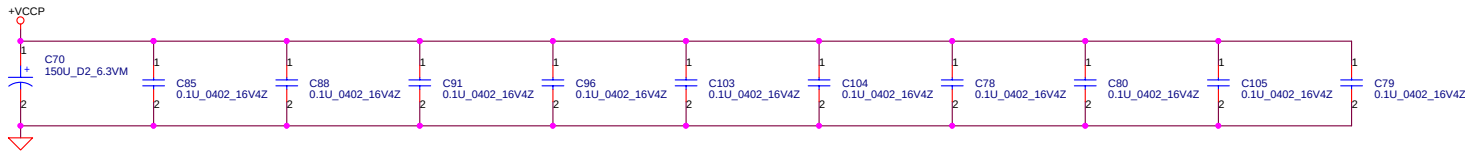
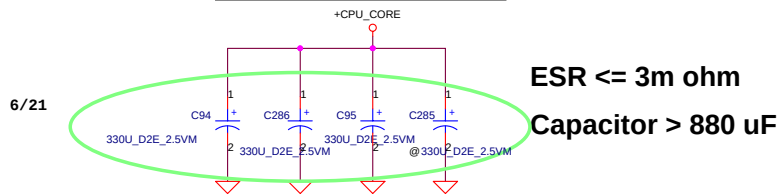
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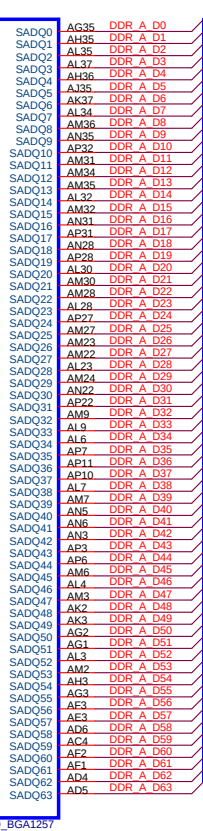
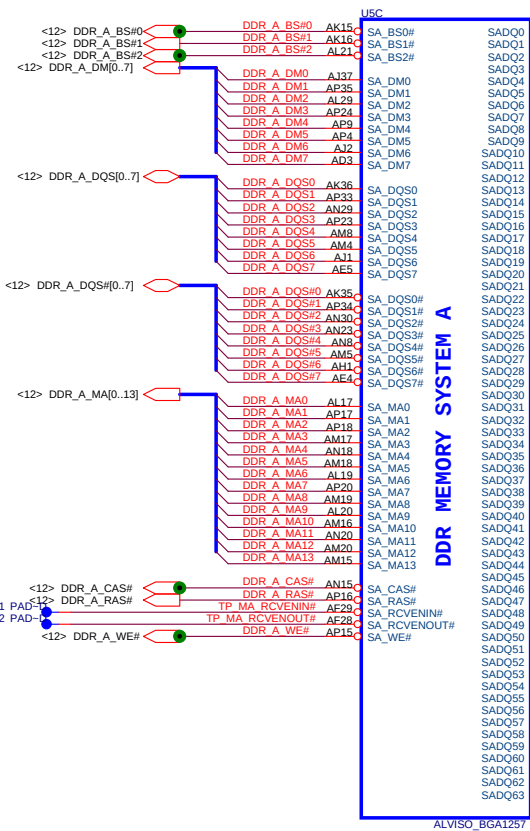




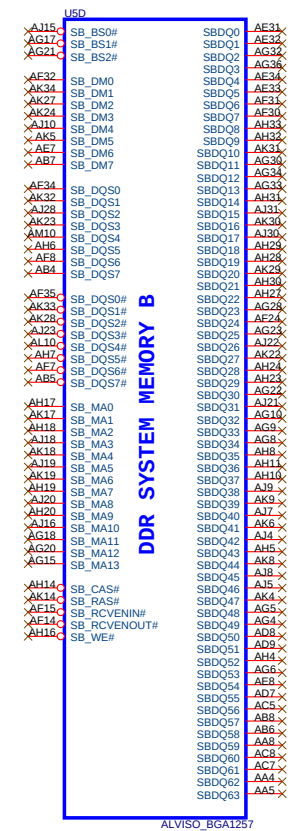


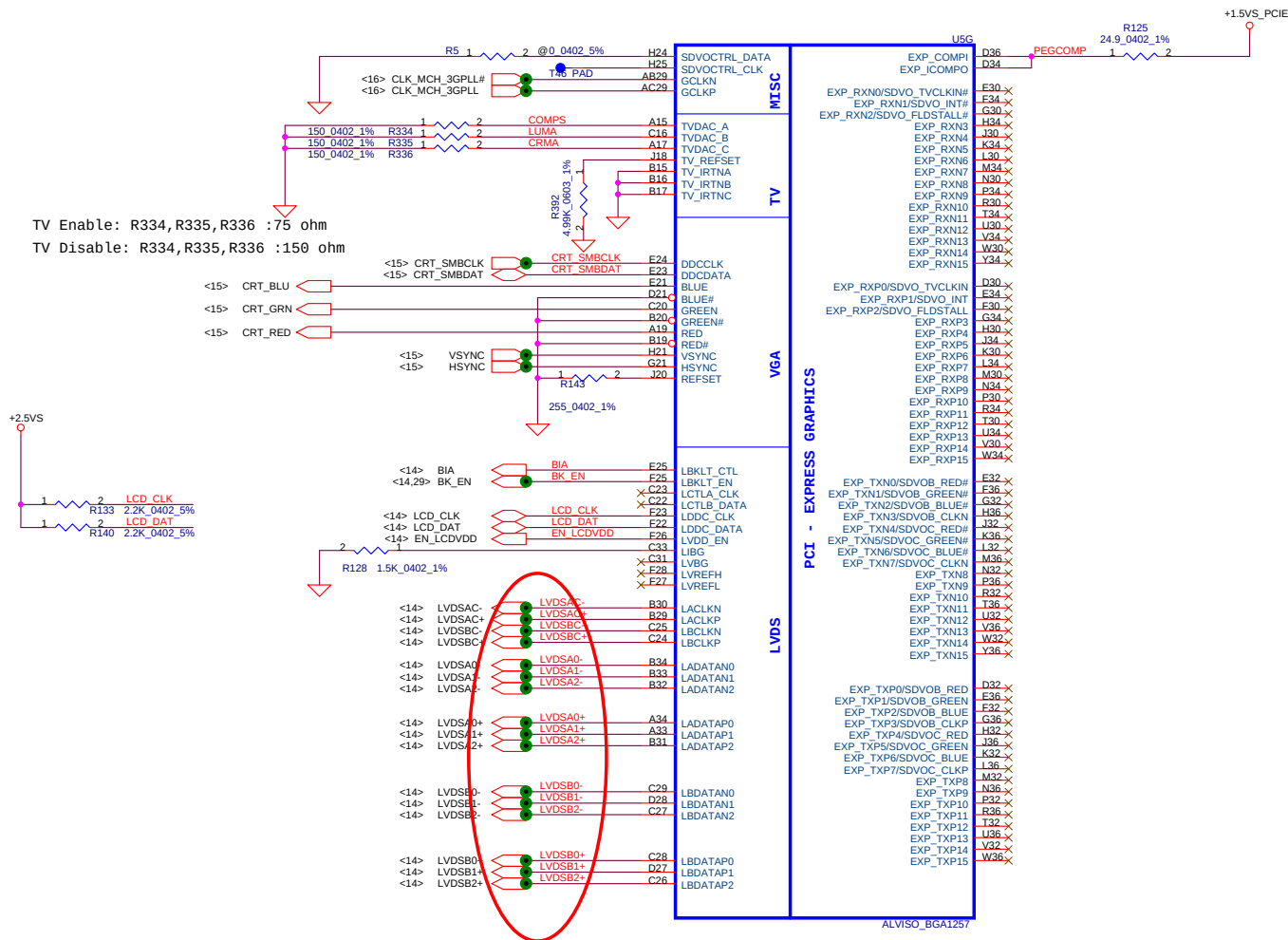
Near VCORE regulator.





This Symbol as same as Intel CRB schematic. So Layout Guide will show these signals routed differentially.



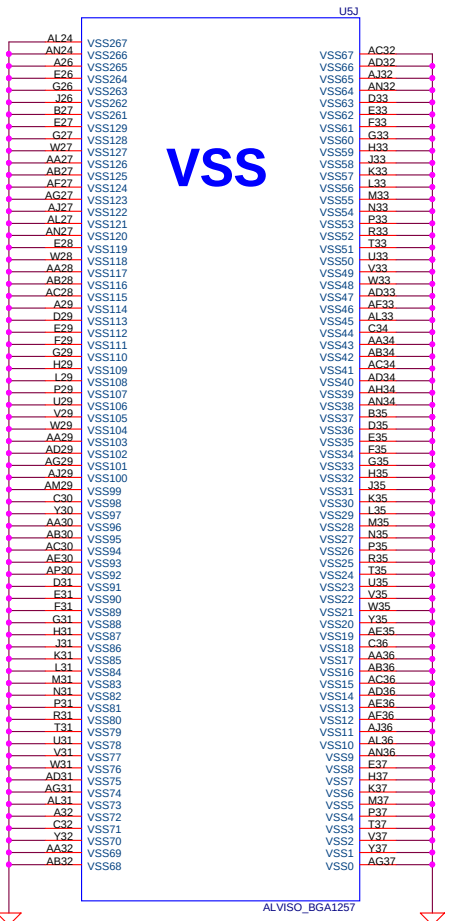
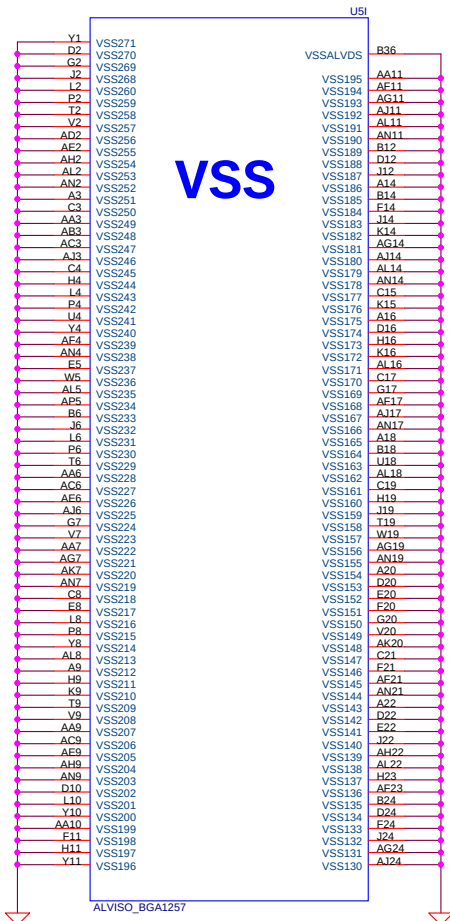
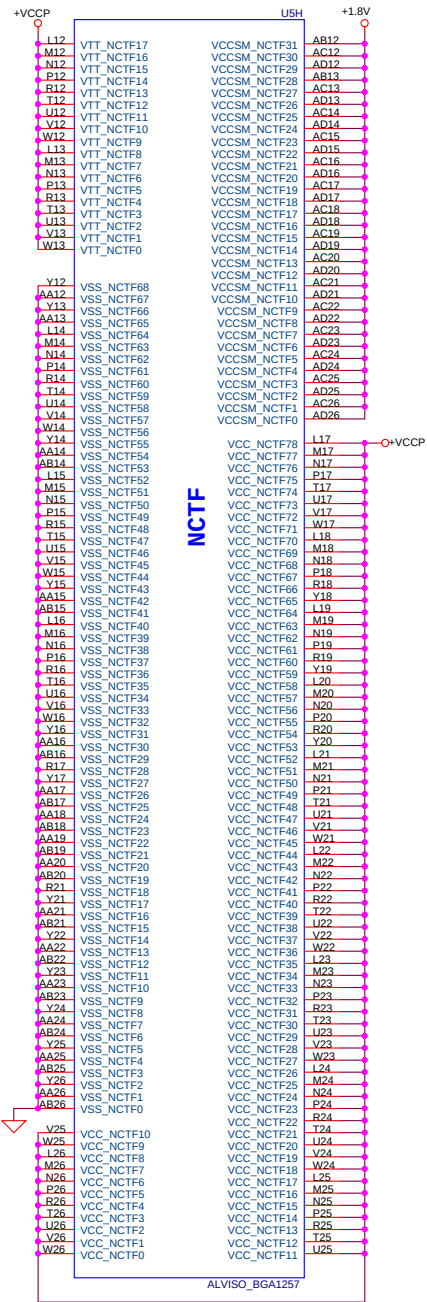


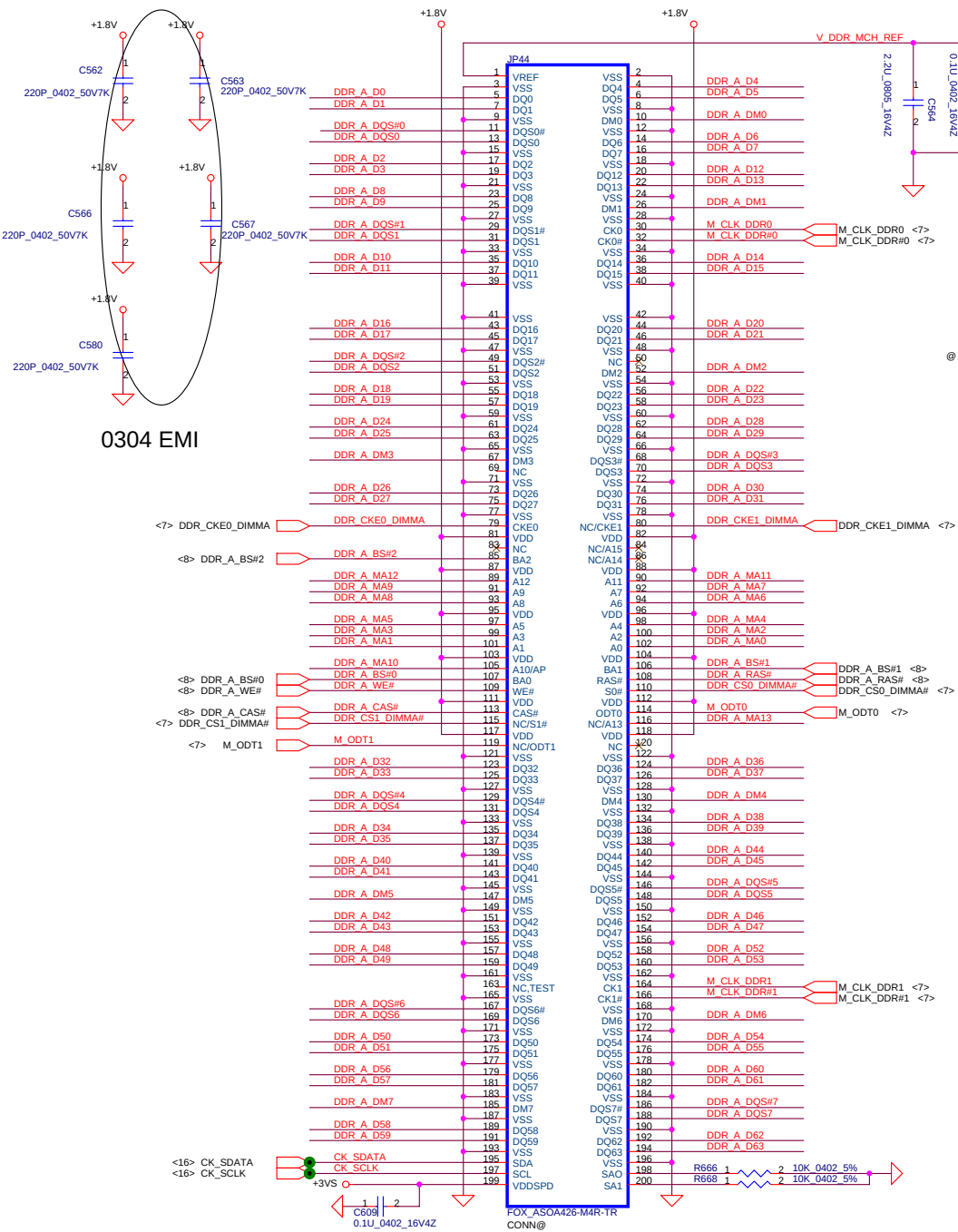
PCI - EXPRESS GRAPHICS

ALVISO_BGA1257

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Alviso(3 of 5)	
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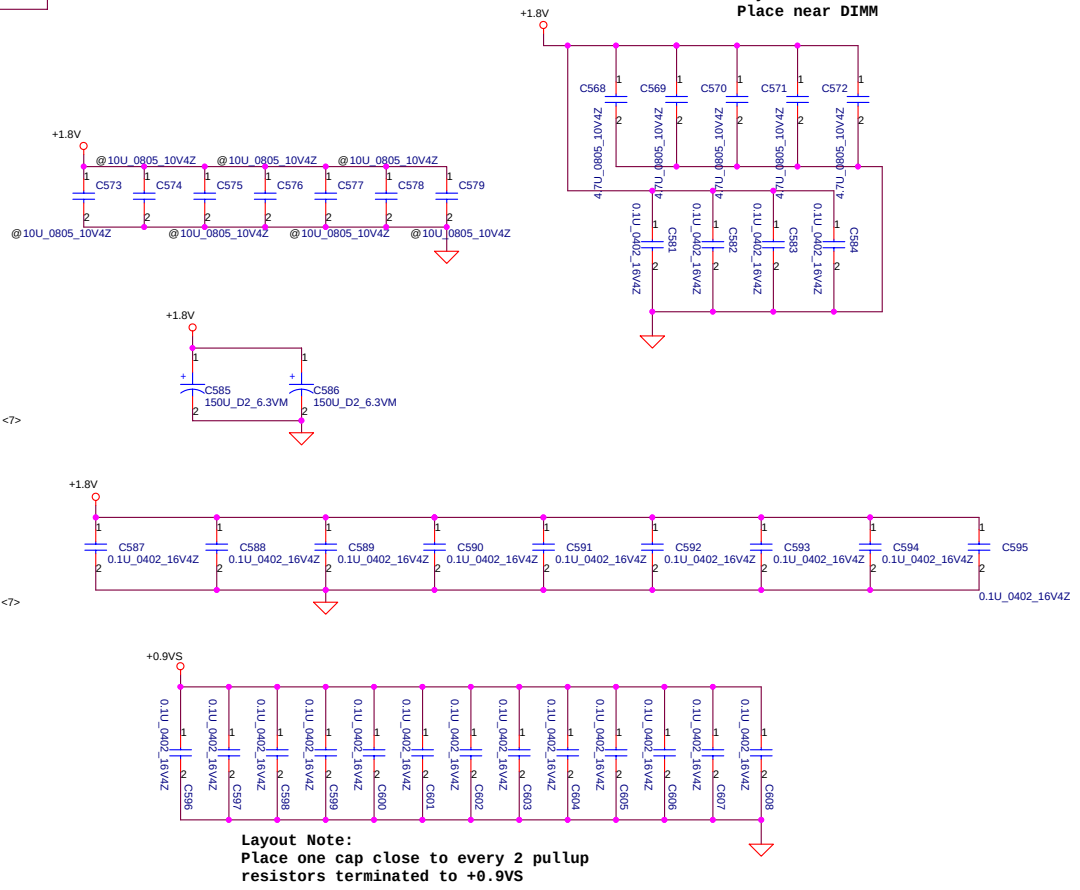
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- DDR_A_D[0..63] ⇨ DDR A D[0..63]
- DDR_A_DM[0..7] ⇨ DDR A DM[0..7]
- DDR_A_DQS[0..7] ⇨ DDR A DQS[0..7]
- DDR_A_MA[0..13] ⇨ DDR A MA[0..13]
- DDR_A_DQS#0..7 ⇨ DDR A DQS#0..7

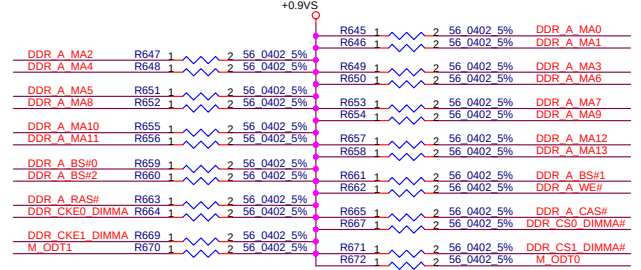
Layout Note:
Place near DIMM

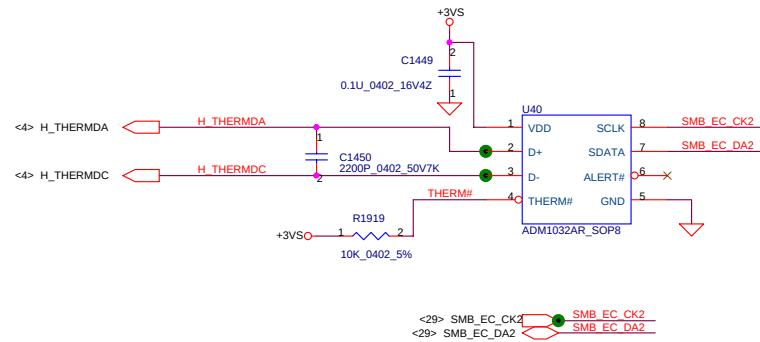
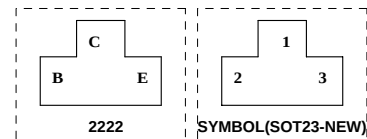


Layout Note:
Place one cap close to every 2 pullup
resistors terminated to +0.9VS

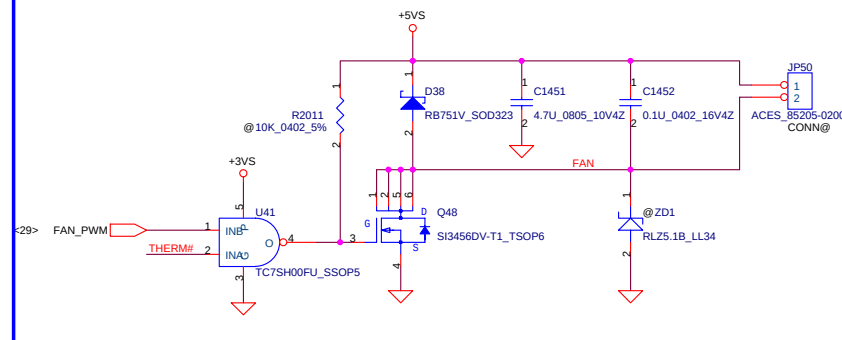
Layout Note:
Place these resistor
closely DIMM0,all
trace length<750 mil

Layout Note:
Place these resistor
closely DIMM0,all
trace length
Max=1.3"



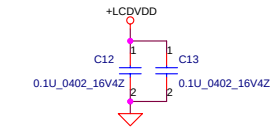


PWM Fan Control circuit

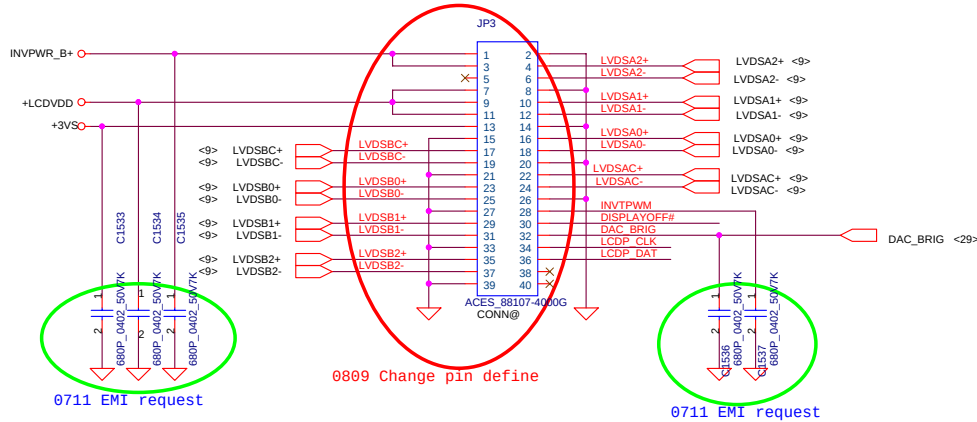


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Title	Thermal sensor and Fan		
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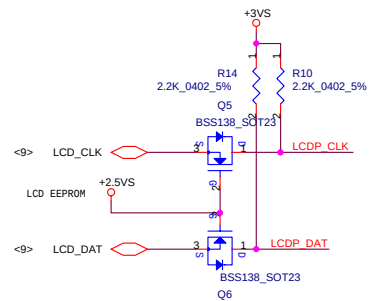
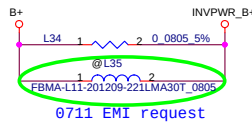
LVDS connector



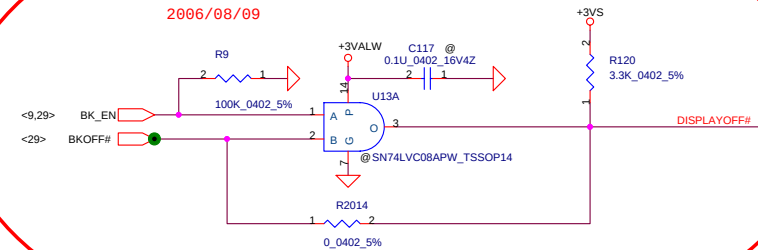
0809 Change pin define

0711 EMI request

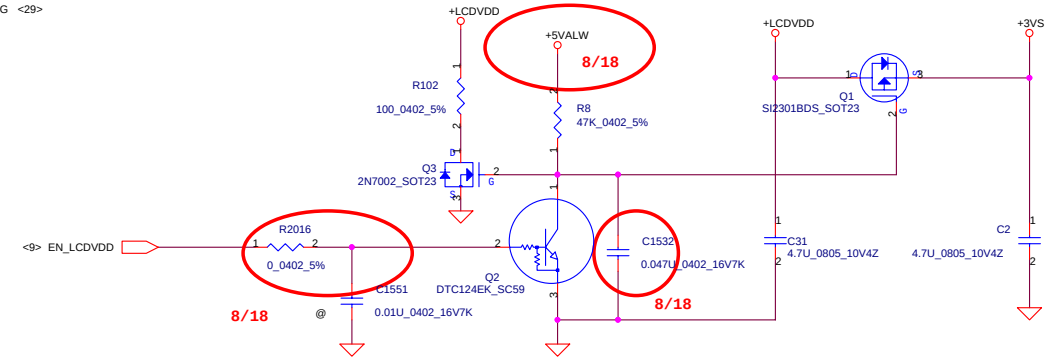
0711 EMI request



2006/08/09

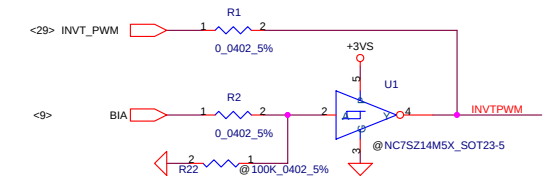


Aviso LCD/PANEL BD. CONN.



8/18

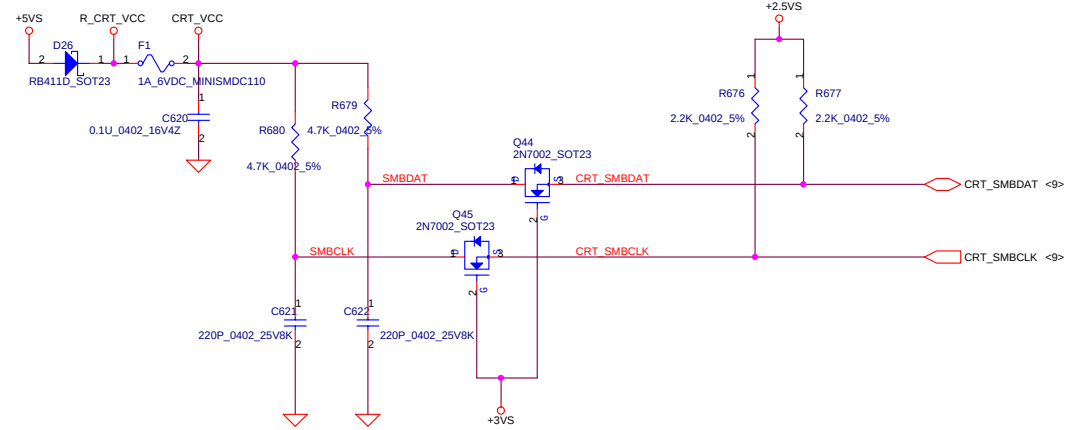
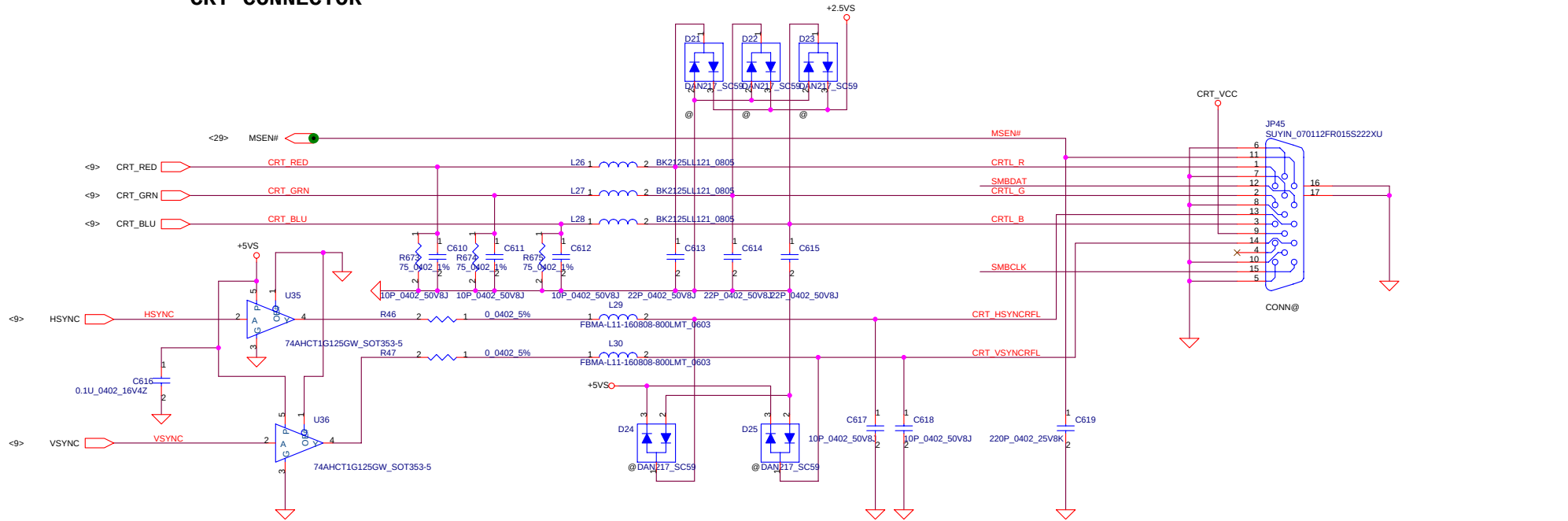
8/18

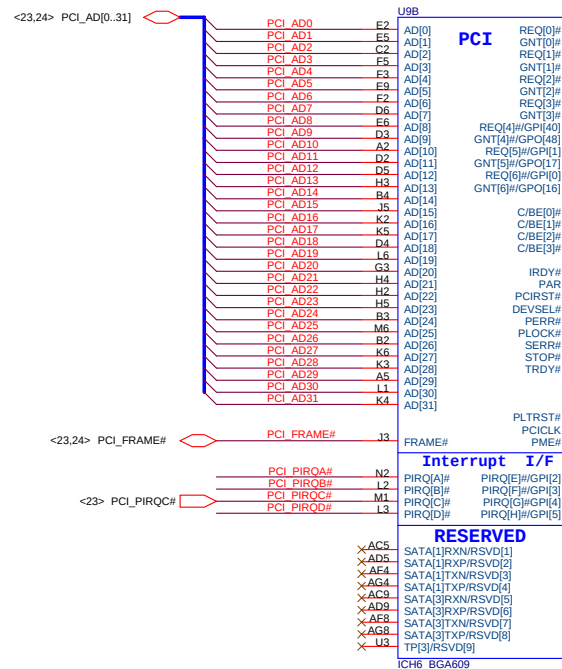


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Title	LVDS connector		
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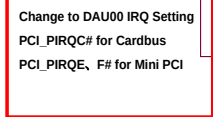
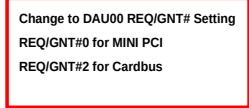
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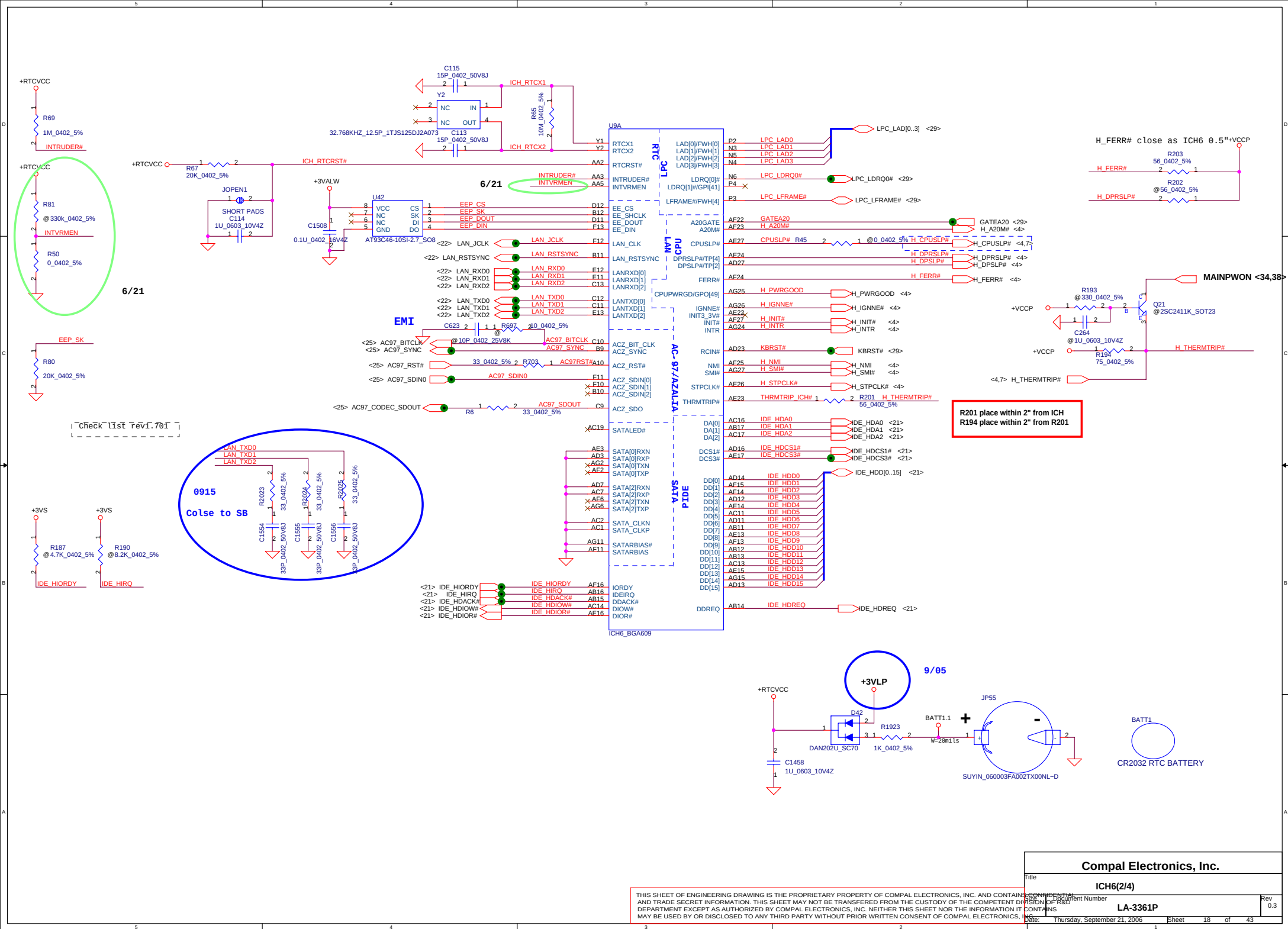
CRT CONNECTOR

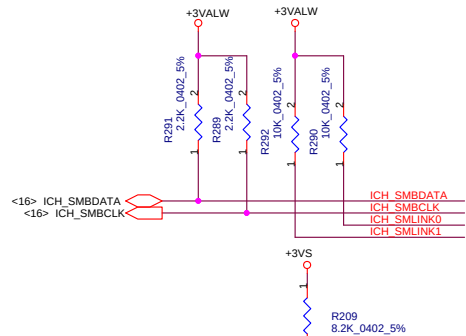




SA8280108G0 S IC 030 FW82801FBM B2 BGA 609P ICH6-M ->
SA8280108E0(R3)/SA8280108D0(R1) S IC 030 FW82801FBM QS ICH6-M BGA 609P



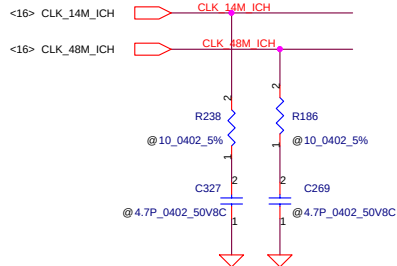




Reserve
Inform BIOS team

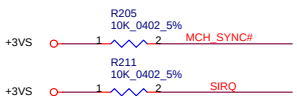
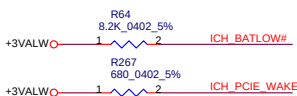
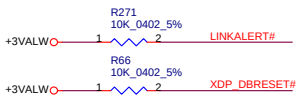
Requires a PU Resistor to Vcc3_3(CRB uses 8.2K to Vcc3_3)
CLK RUN no work to pull down

5/30

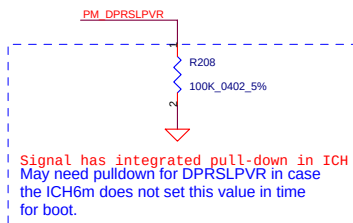
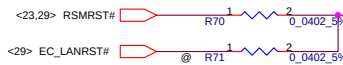


ICH6 VER1.5 GPIO12 +3VS plan

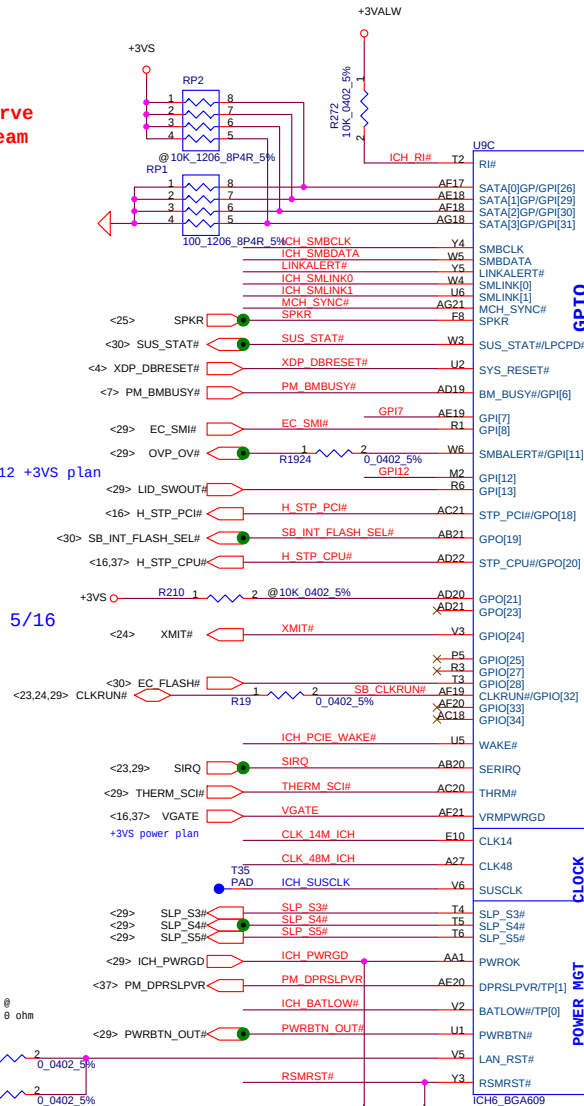
GPIO 23 PLTRST_VGA# Delete 5/16



8.2 k pull-up to Vcc3_3(CRB uses 10 k)



Signal has integrated pull-down in ICH
May need pulldown for DPRSLPVR in case
the ICH6m does not set this value in time
for boot.



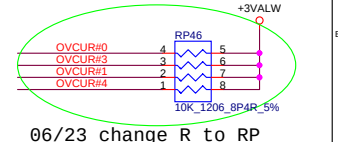
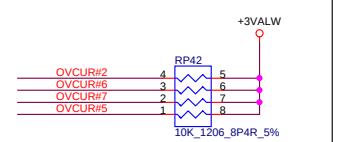
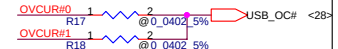
PCI-EXPRESS

DIRECT MEDIA INTERFACE

CLOCK

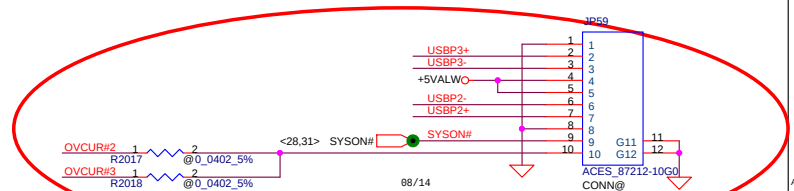
USB

POWER MGT

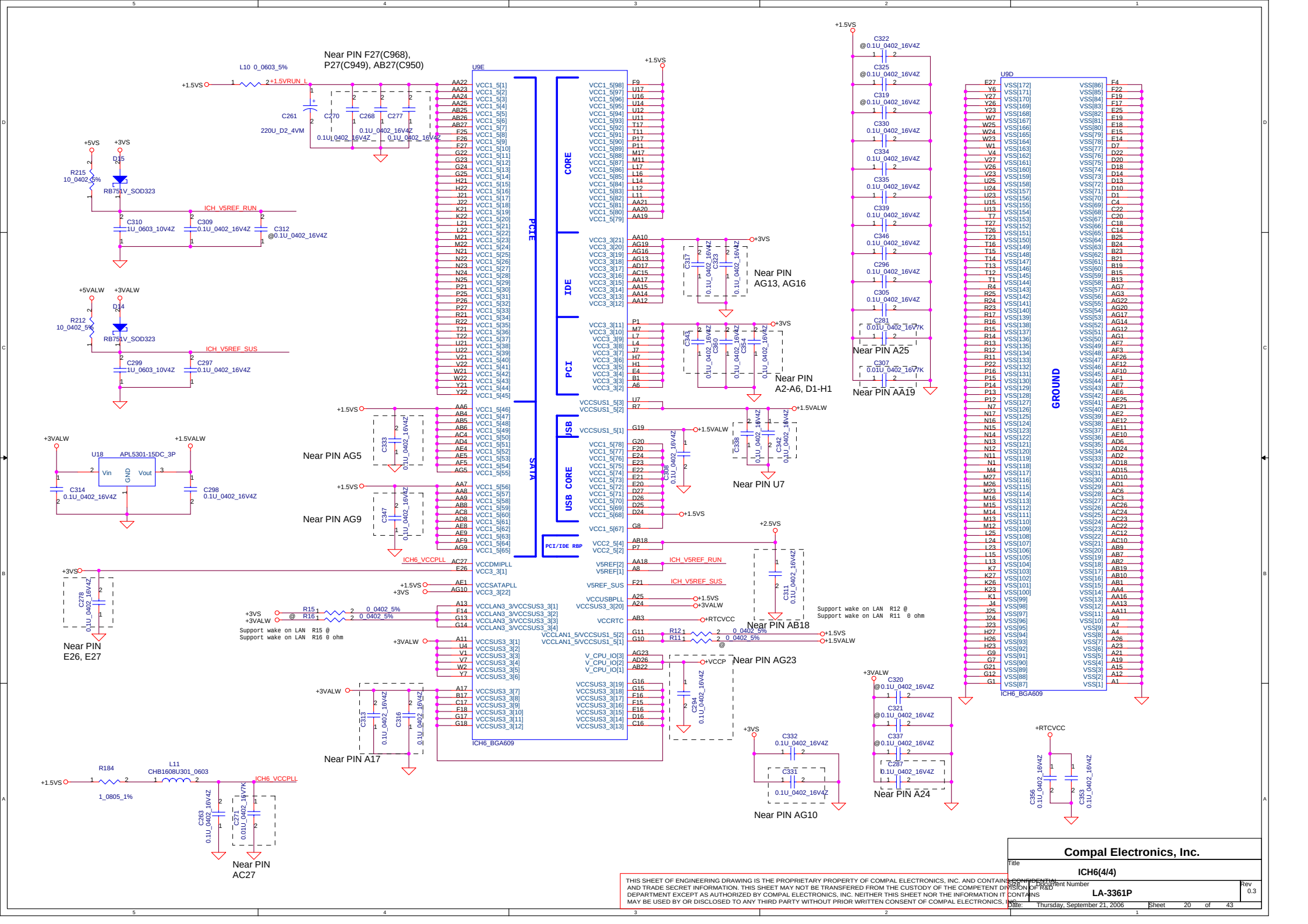


06/23 change R to RP

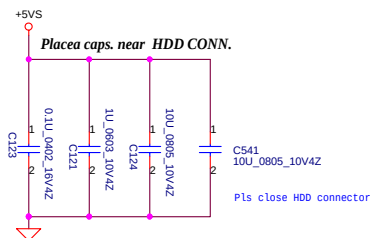
USBPN/USBPP impedance 45 Ohm



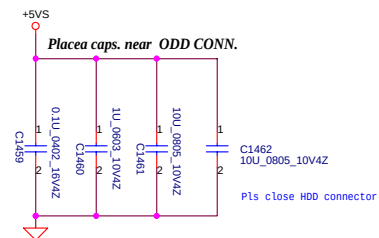
08/10 Add two USB port to subboard



IDE_HDD[0..15] <18>

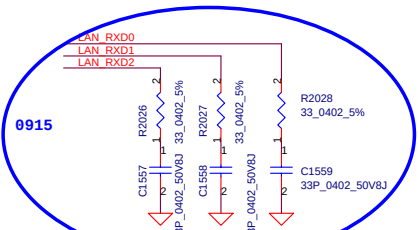


Pull high set slave



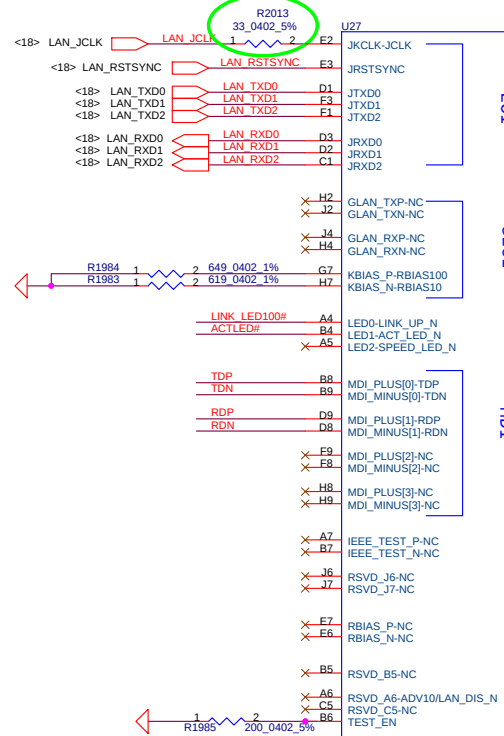
06/23

R27
IDE_ACT# 1 2
10K 0402 5%

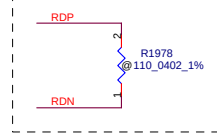


close to LAN chipset

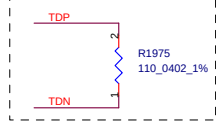
0718 Intel checklist recommend



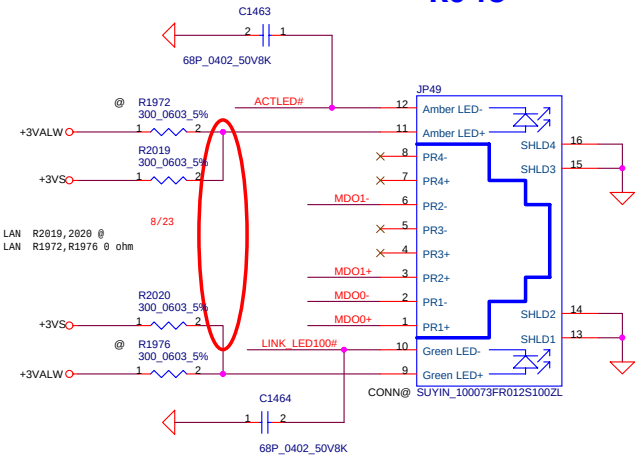
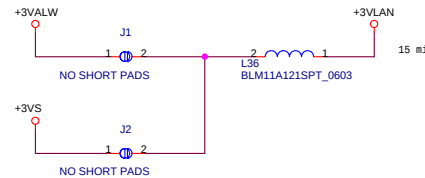
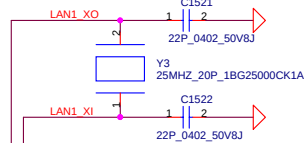
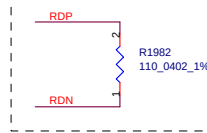
close to U25chip(Intel rule)



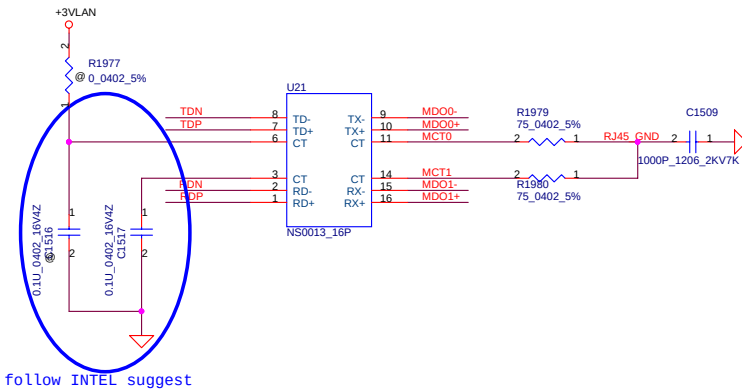
close to U25chip(Intel rule)



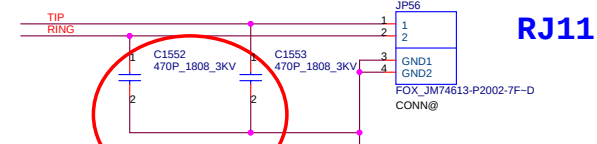
close to U21



RJ45



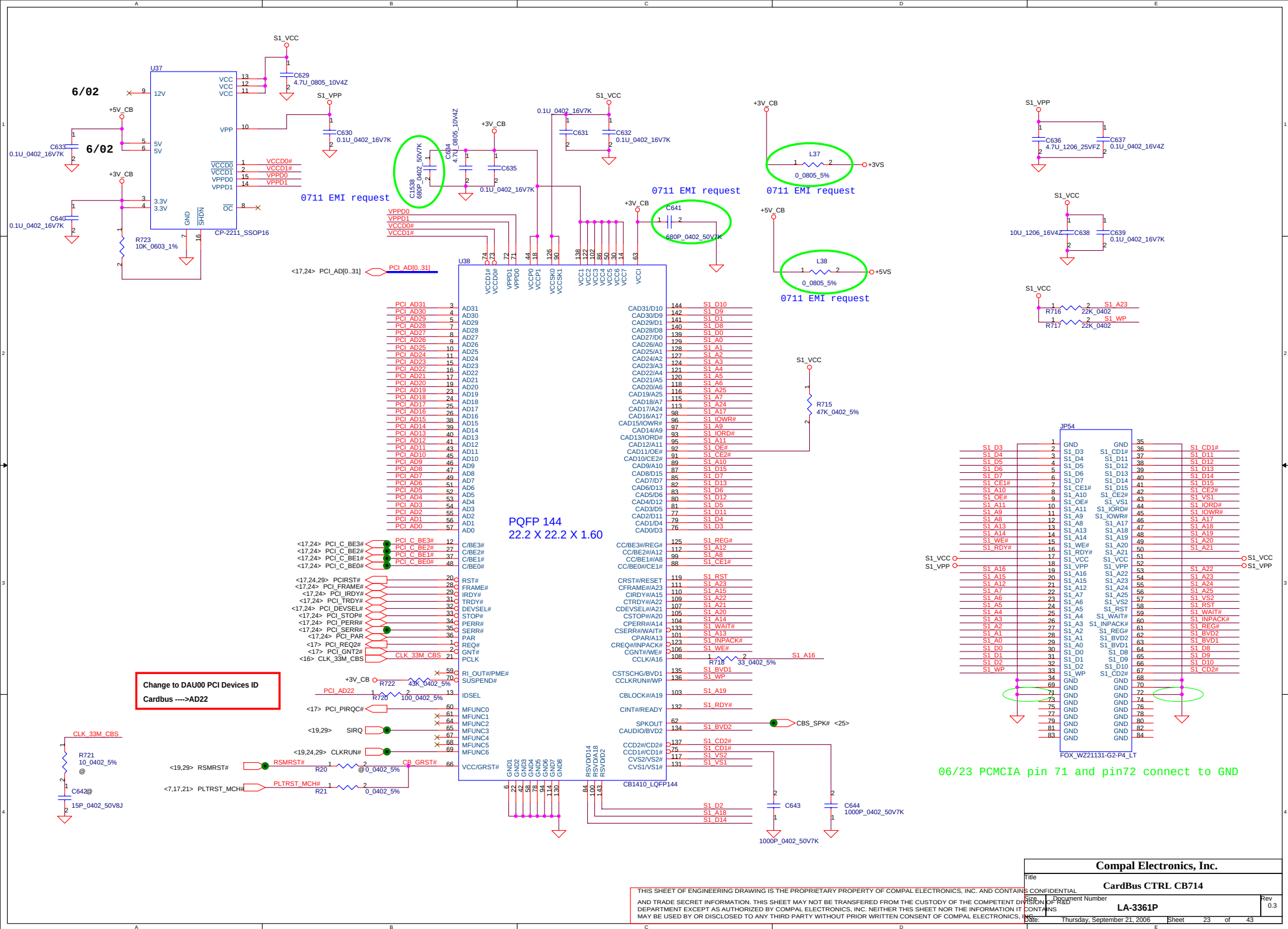
09/15 follow INTEL suggest

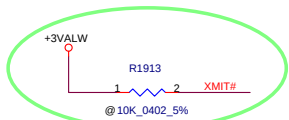


RJ11



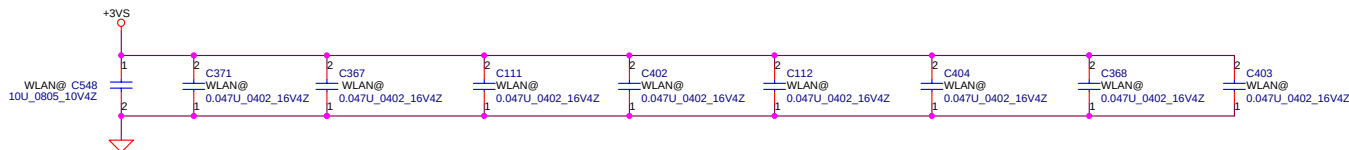
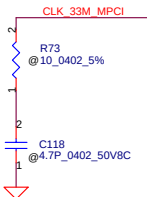
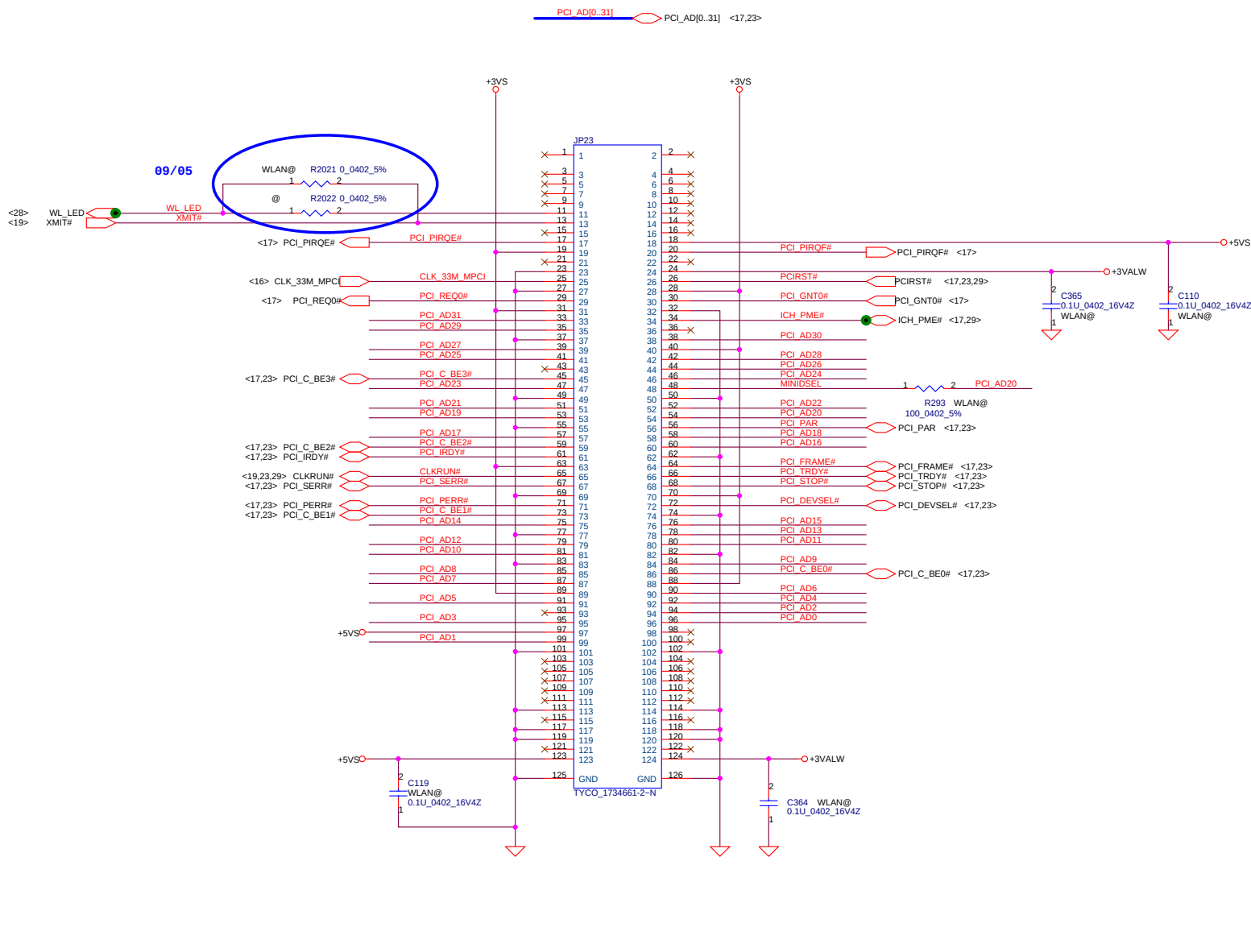
Security Classification				Compal Secret Data				Compal Electronics, Inc.			
Issued Date				2005/03/10				82562EZ LAN			
Deciphered Date				2006/03/10				LA-3361P			
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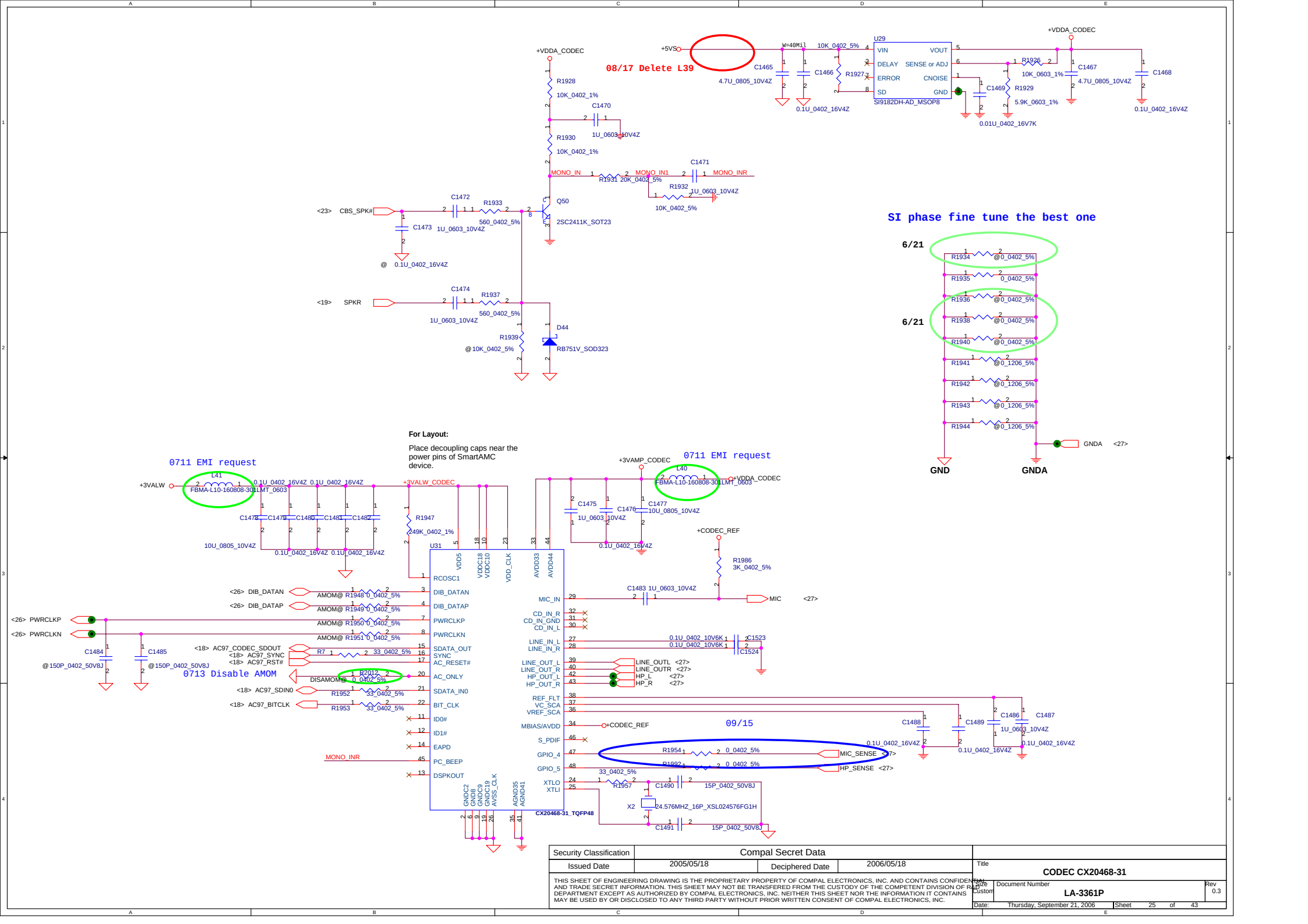


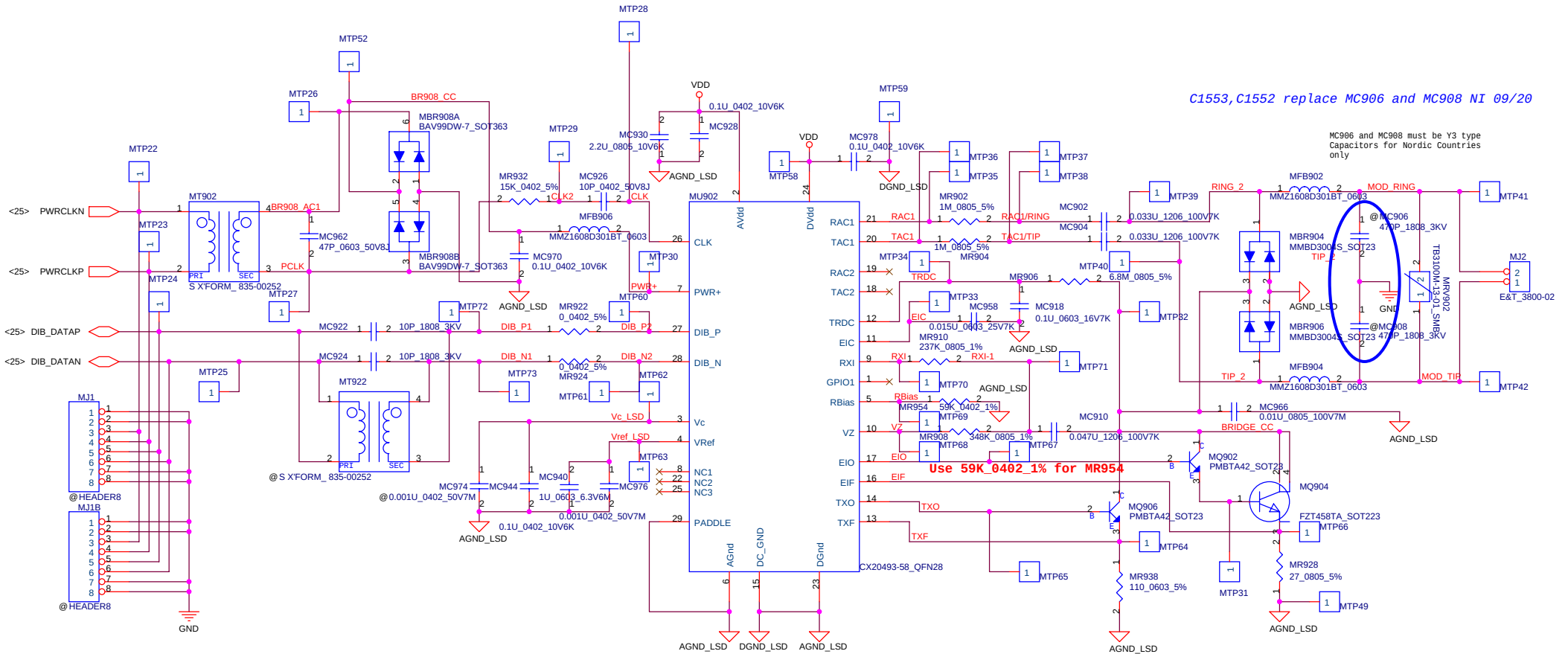


6/23

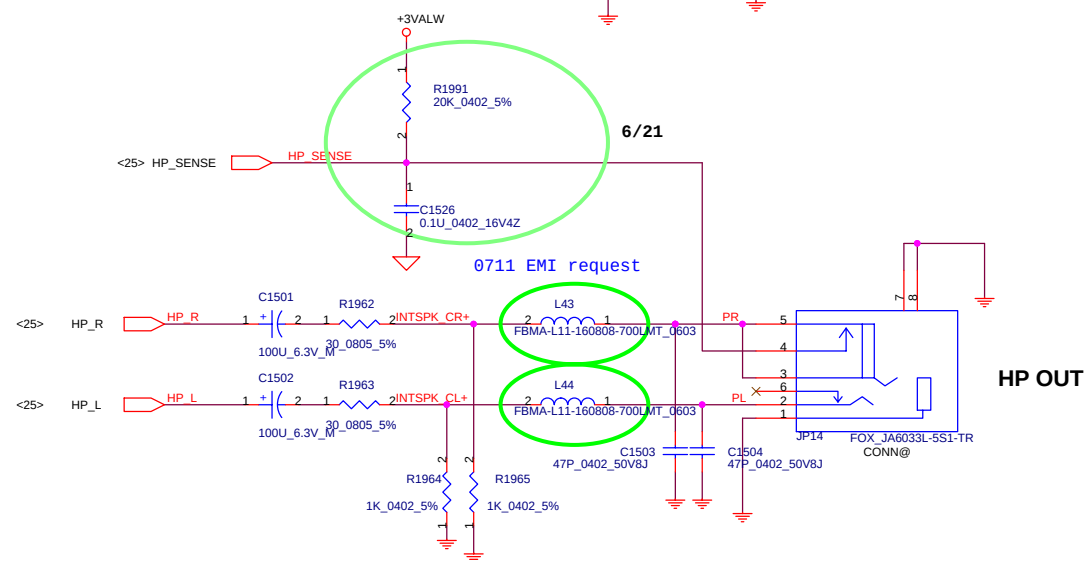
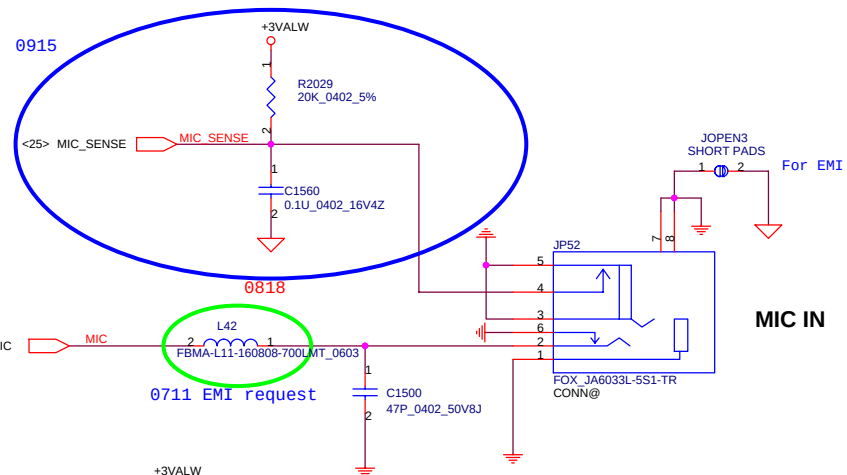
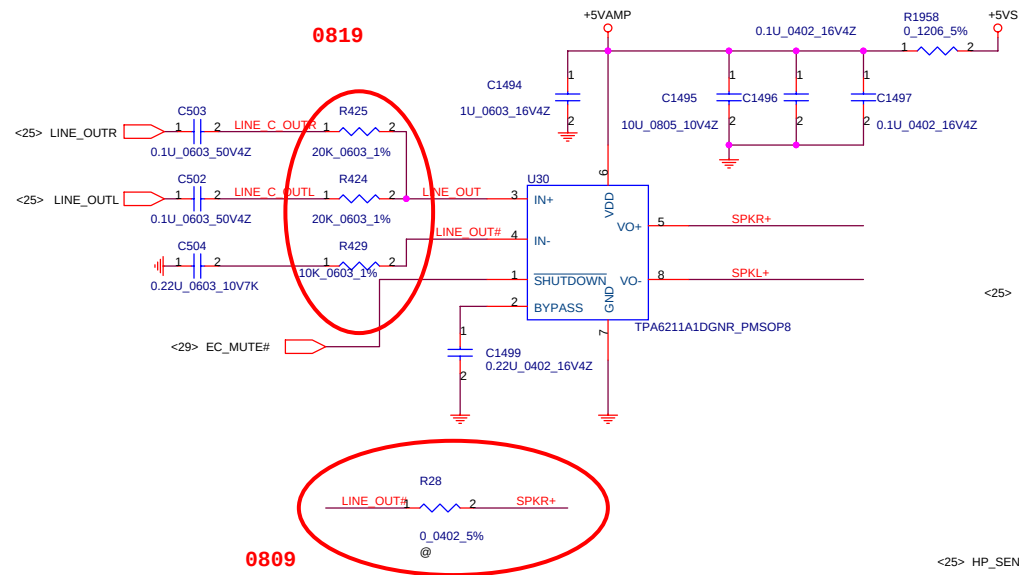
09/05



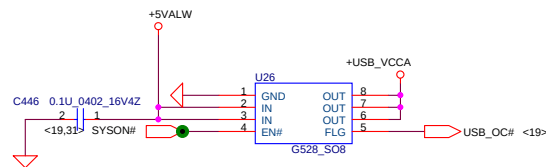




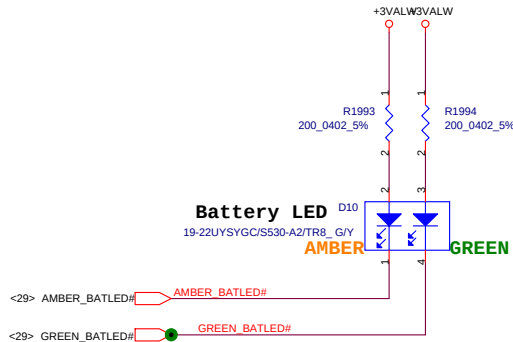
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			LA-3361P	0.3
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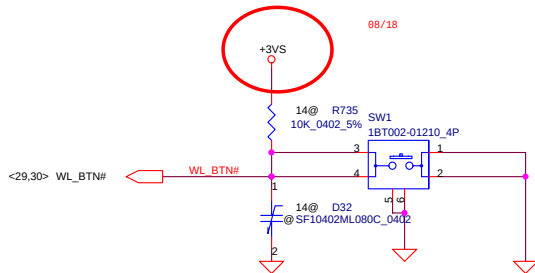
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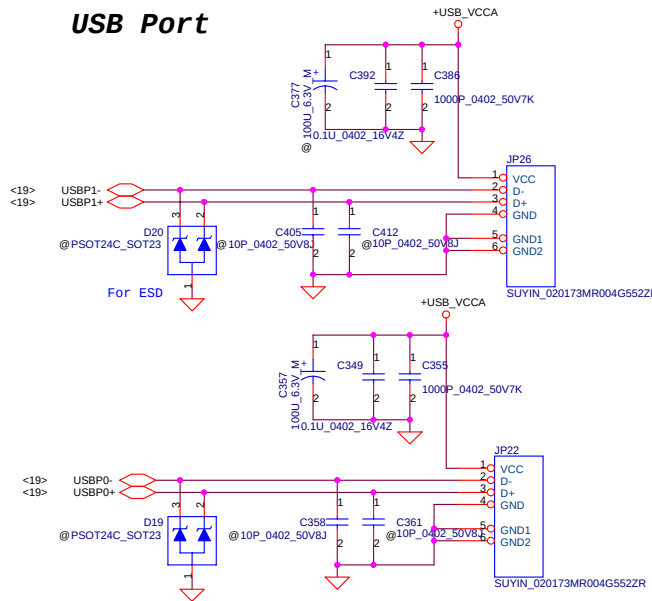
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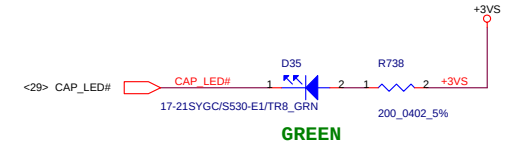
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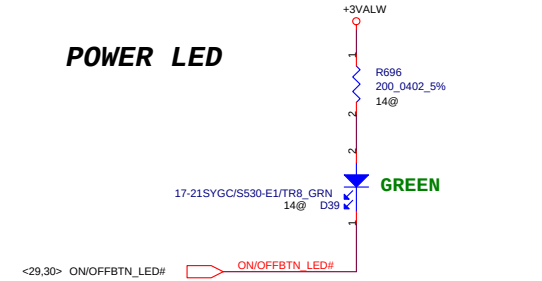
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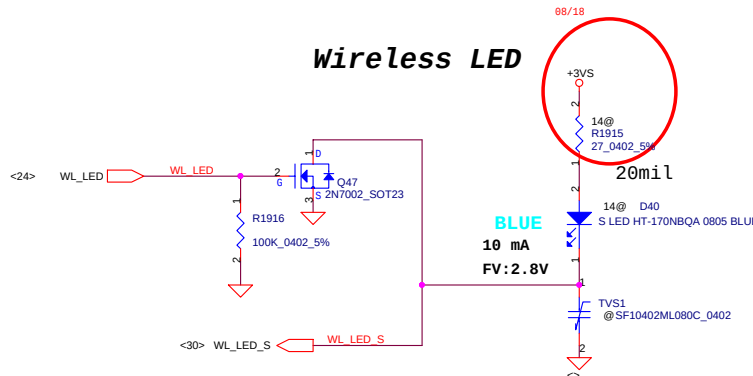
CAP LED



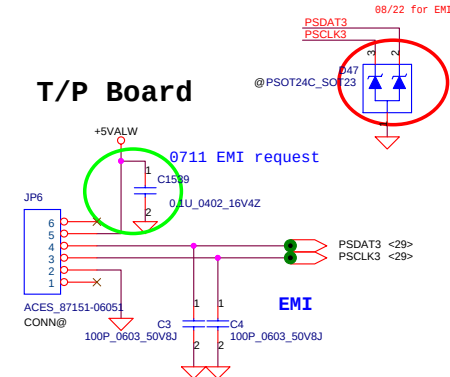
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Wireless LED

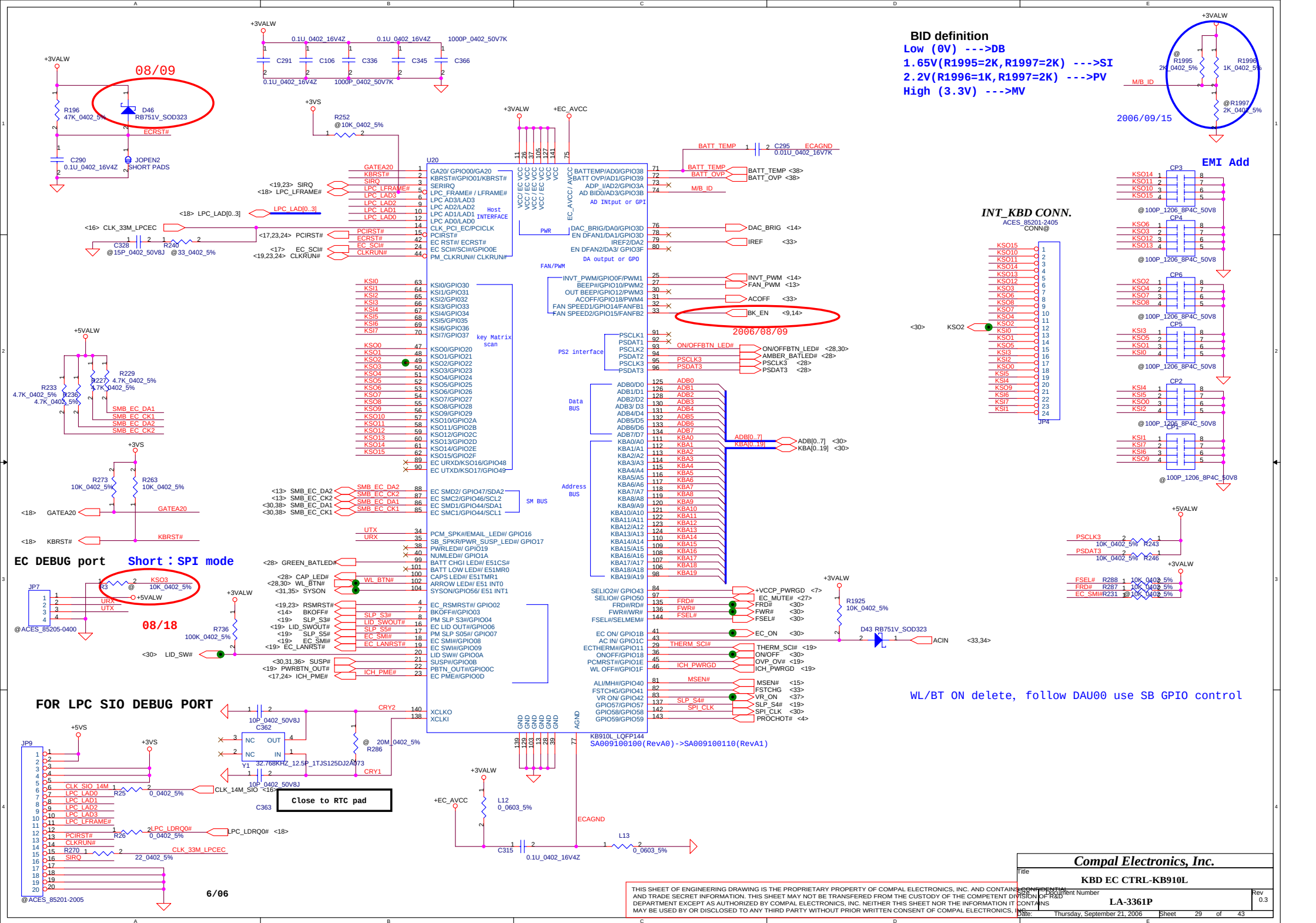


T/P Board



Compal Electronics, Inc.			
Title	USB Port	Document Number	Rev
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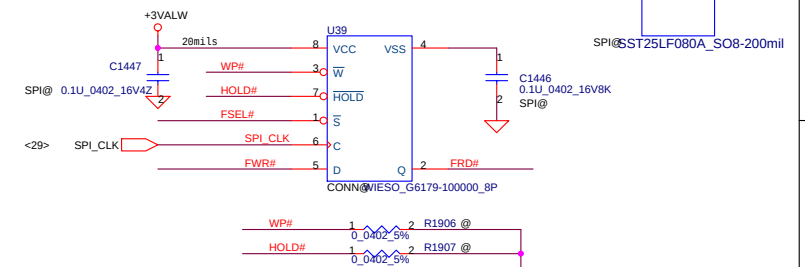
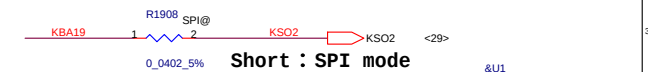
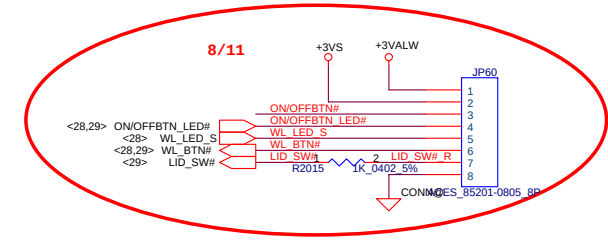
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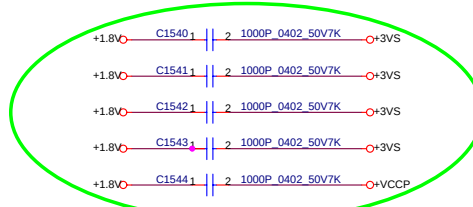
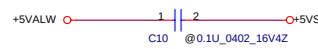
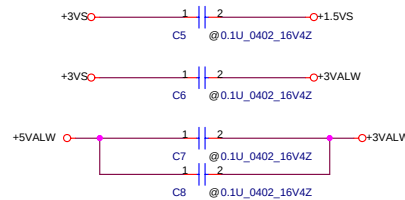
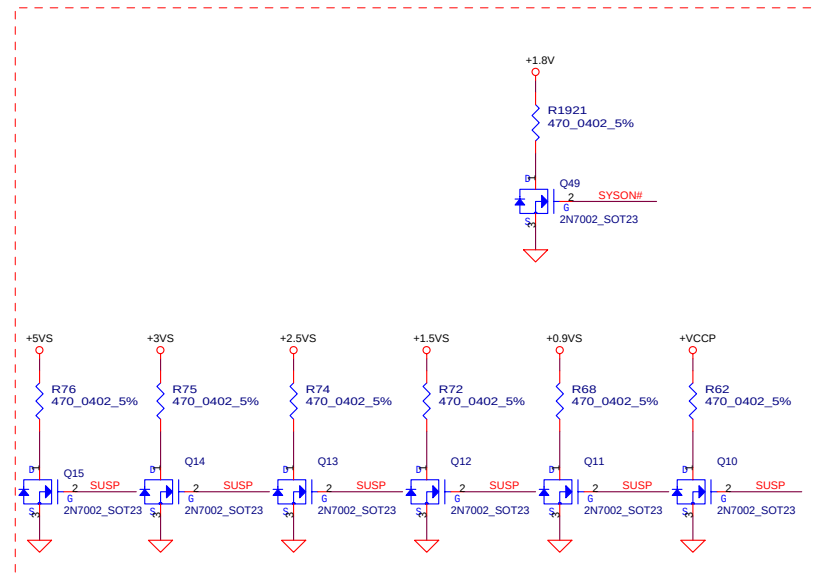
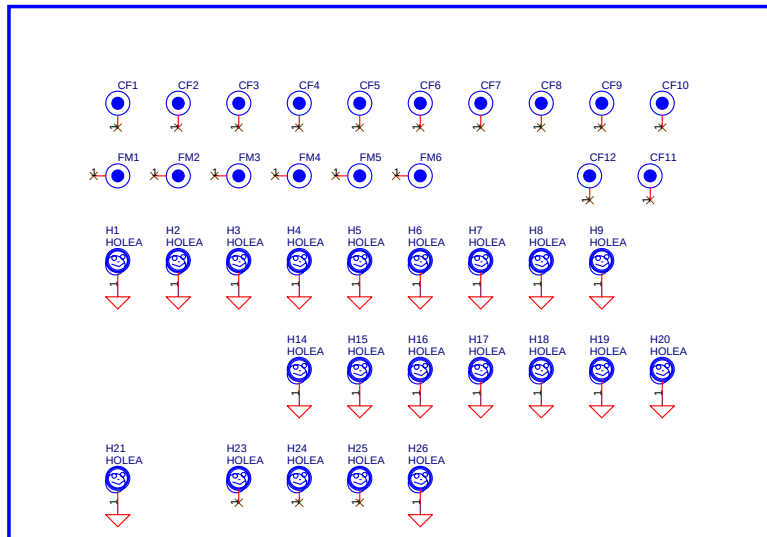
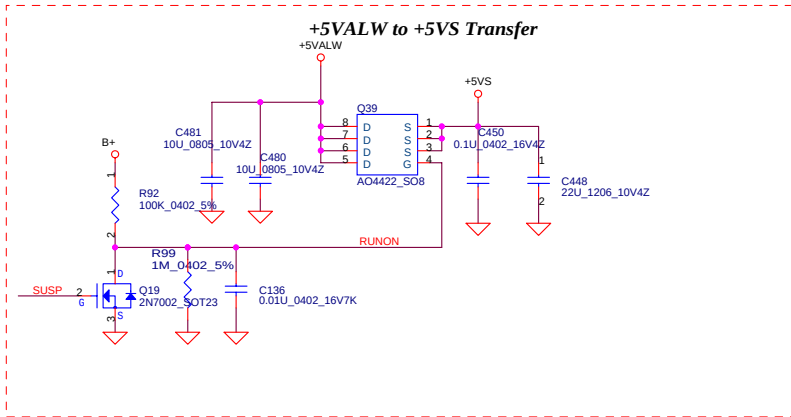
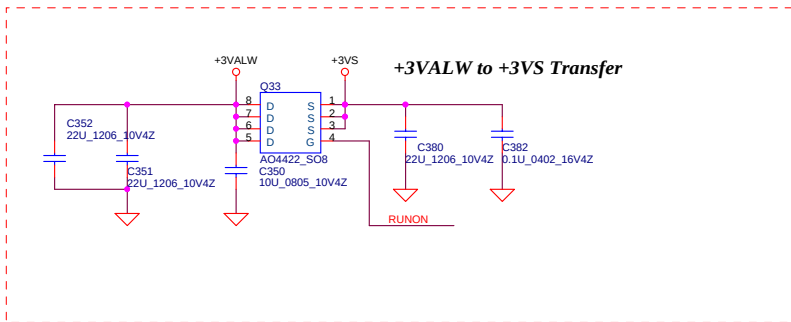


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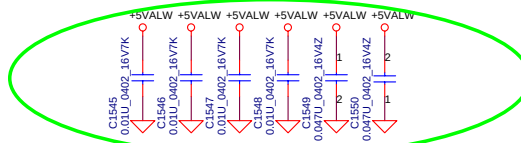
The diagram shows the pin configuration for the SUYIN 80065AR-040G2T microcontroller. The pins are numbered 1 through 40 along the bottom edge. Functions are listed next to each pin number.

Pin Number	Function
1	KBA16
2	KBA15
3	KBA14
4	KBA13
5	KBA12
6	KBA11
7	KBA9
8	KBA8
9	FWR#
10	RESET#1
11	INT_FLASH_EN#
12	INT_FLASH_SEL
13	KBA18
14	KBA7
15	KBA6
16	KBA5
17	KBA4
18	KBA3
19	KBA2
20	KBA1
21	SUYN_80065AR-040G2T
22	KBA17
23	KBA19
24	ADB7
25	ADB6
26	ADB5
27	ADB4
28	+3VSWAL
29	ADB3
30	ADB2
31	ADB1
32	FRD#
33	FSEL#
34	KBA0

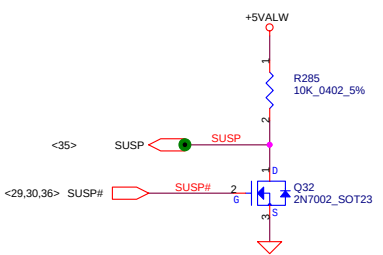
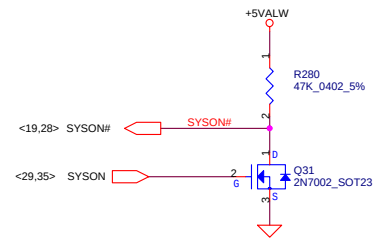
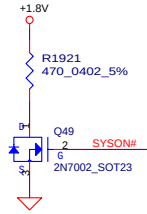




7/11 EMI request

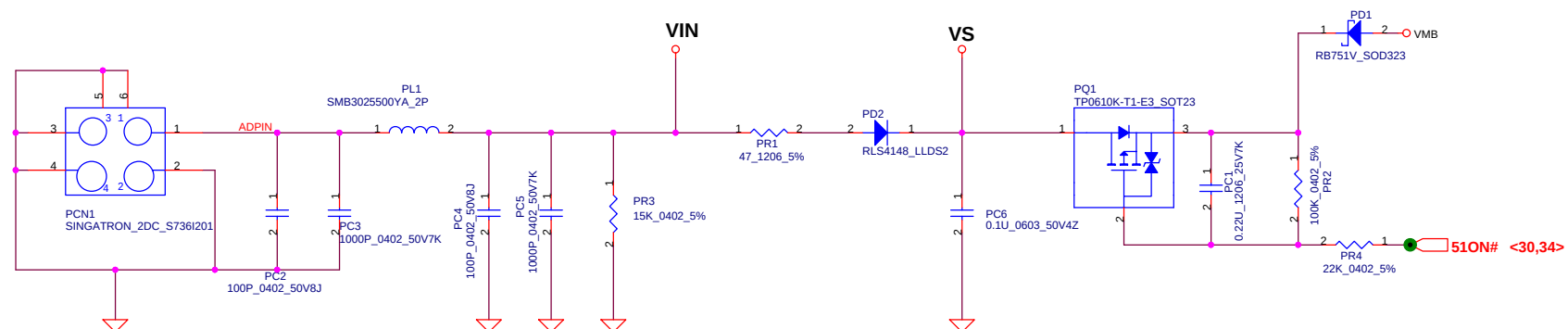


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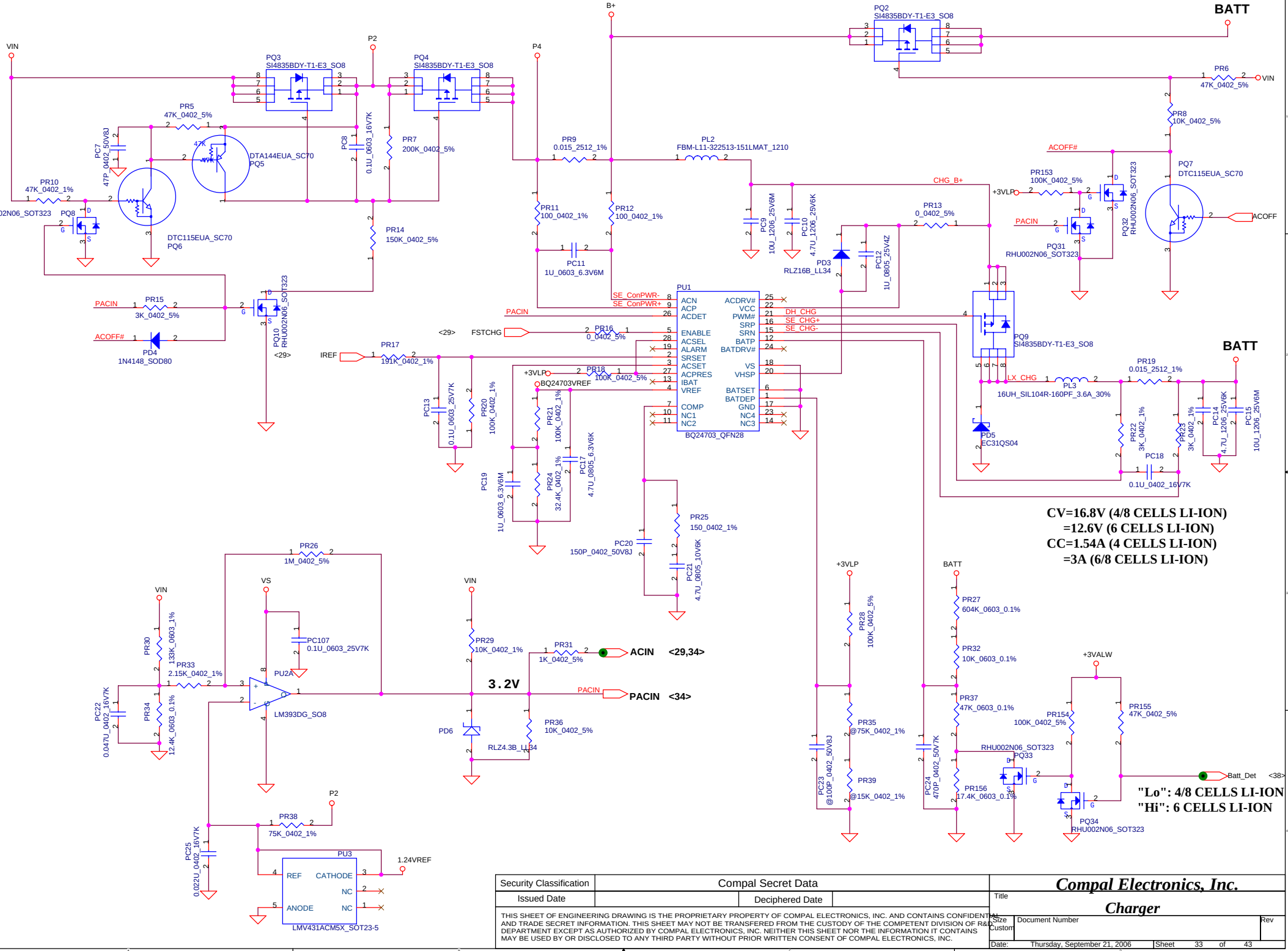


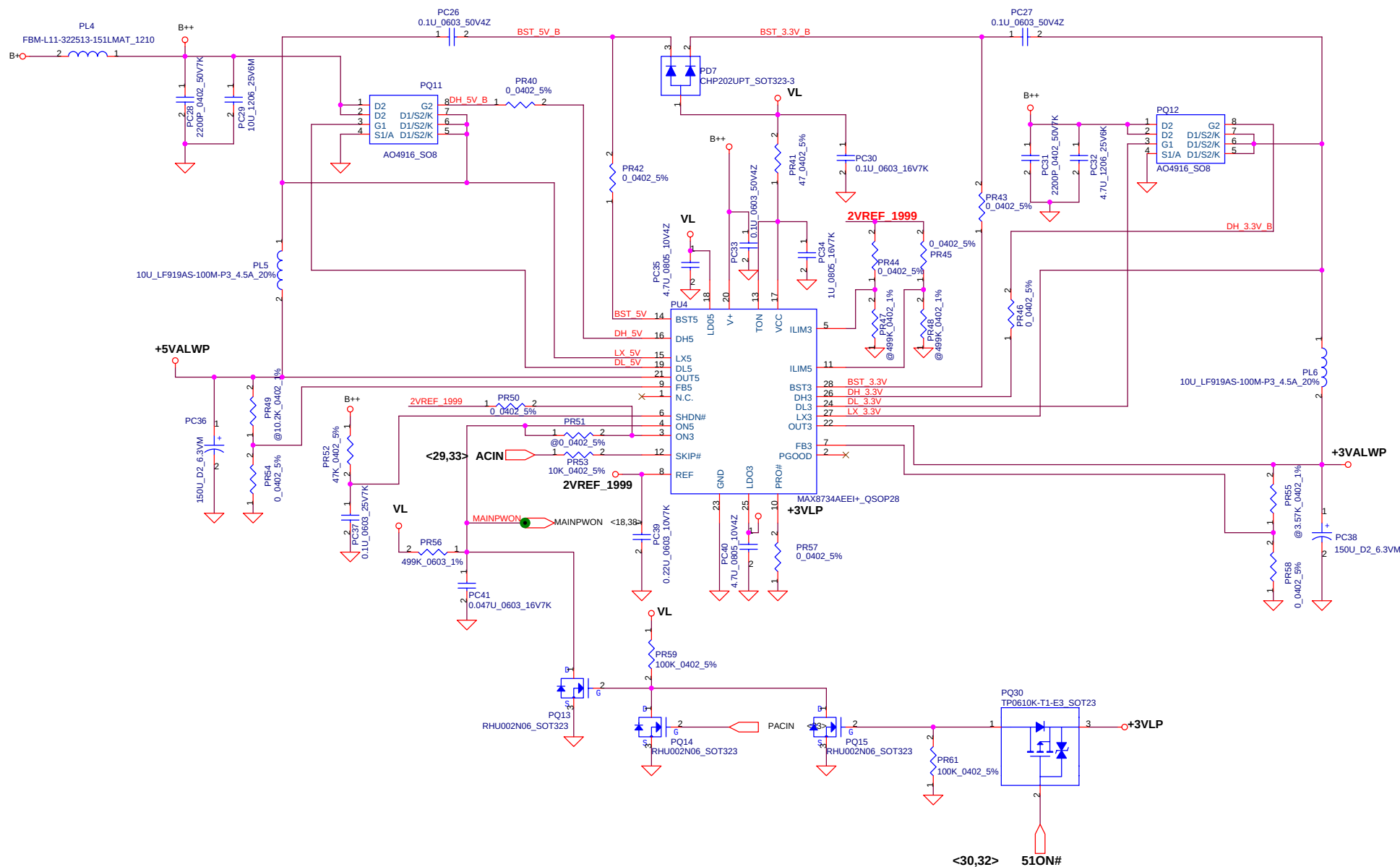
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Title	POWER CONTROL		
Document Number	LA-3361P		
Date	Thursday, September 21, 2006	Sheet	31 of 43

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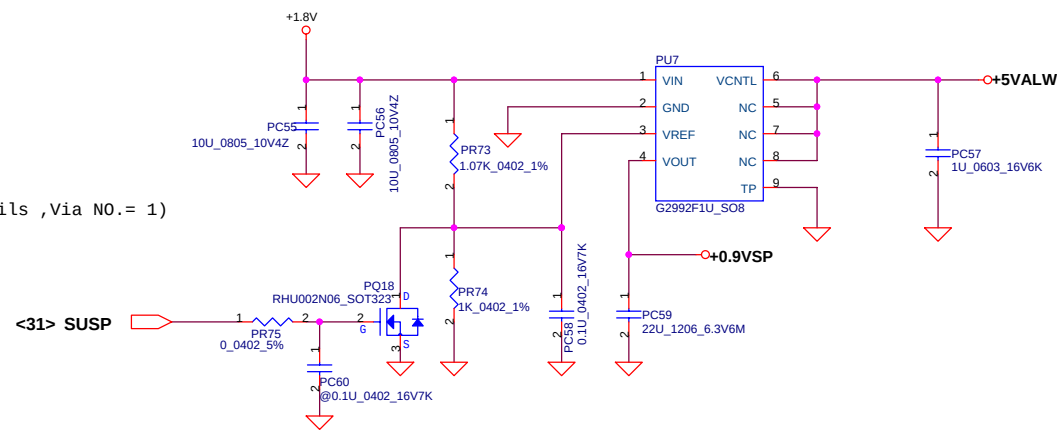
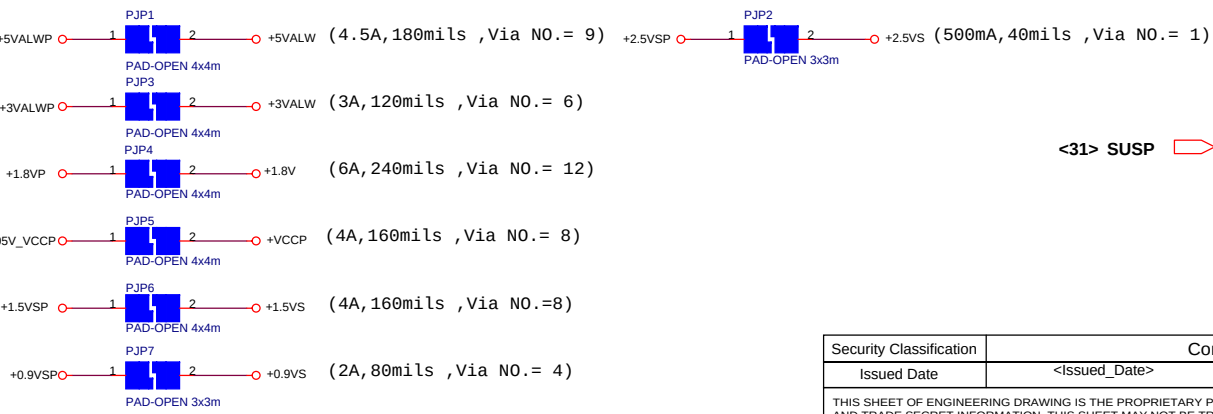
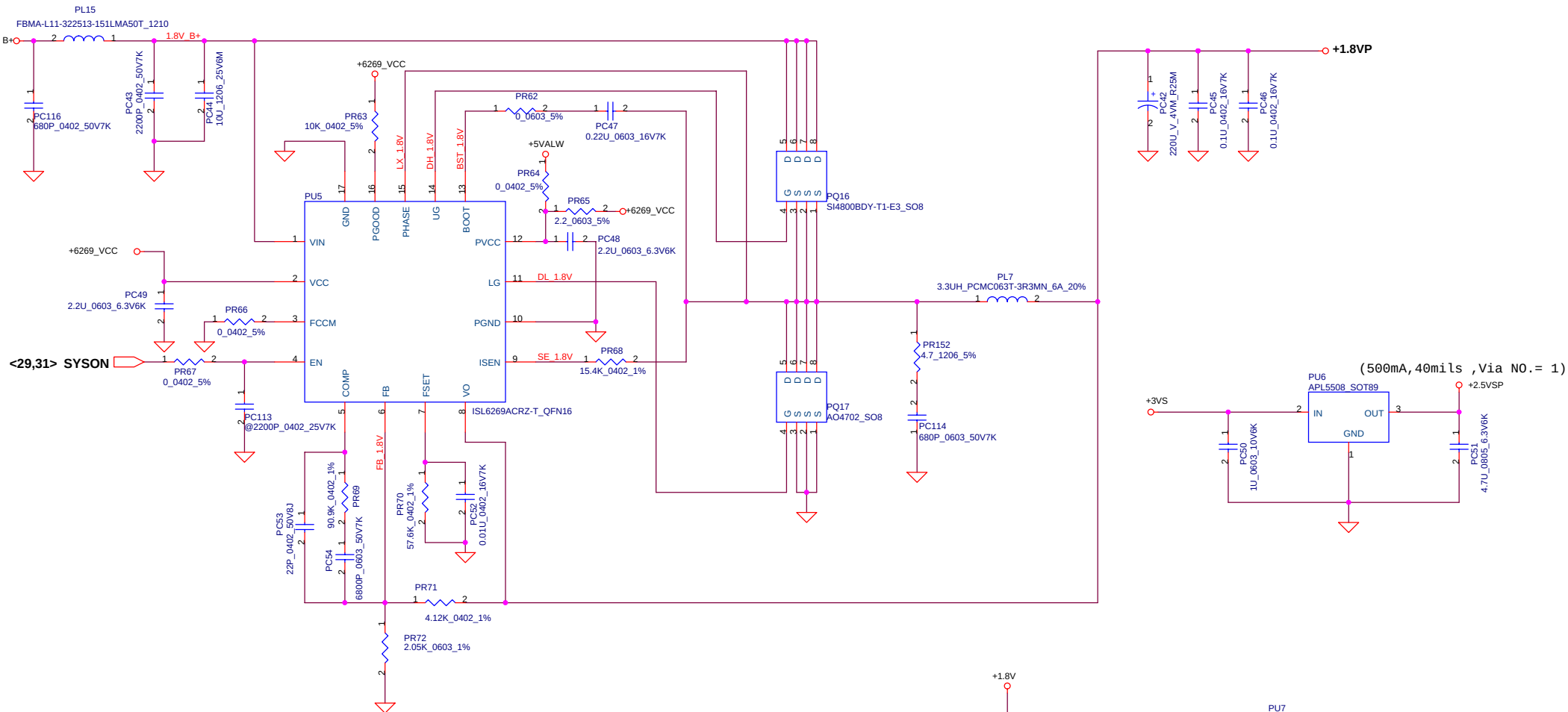


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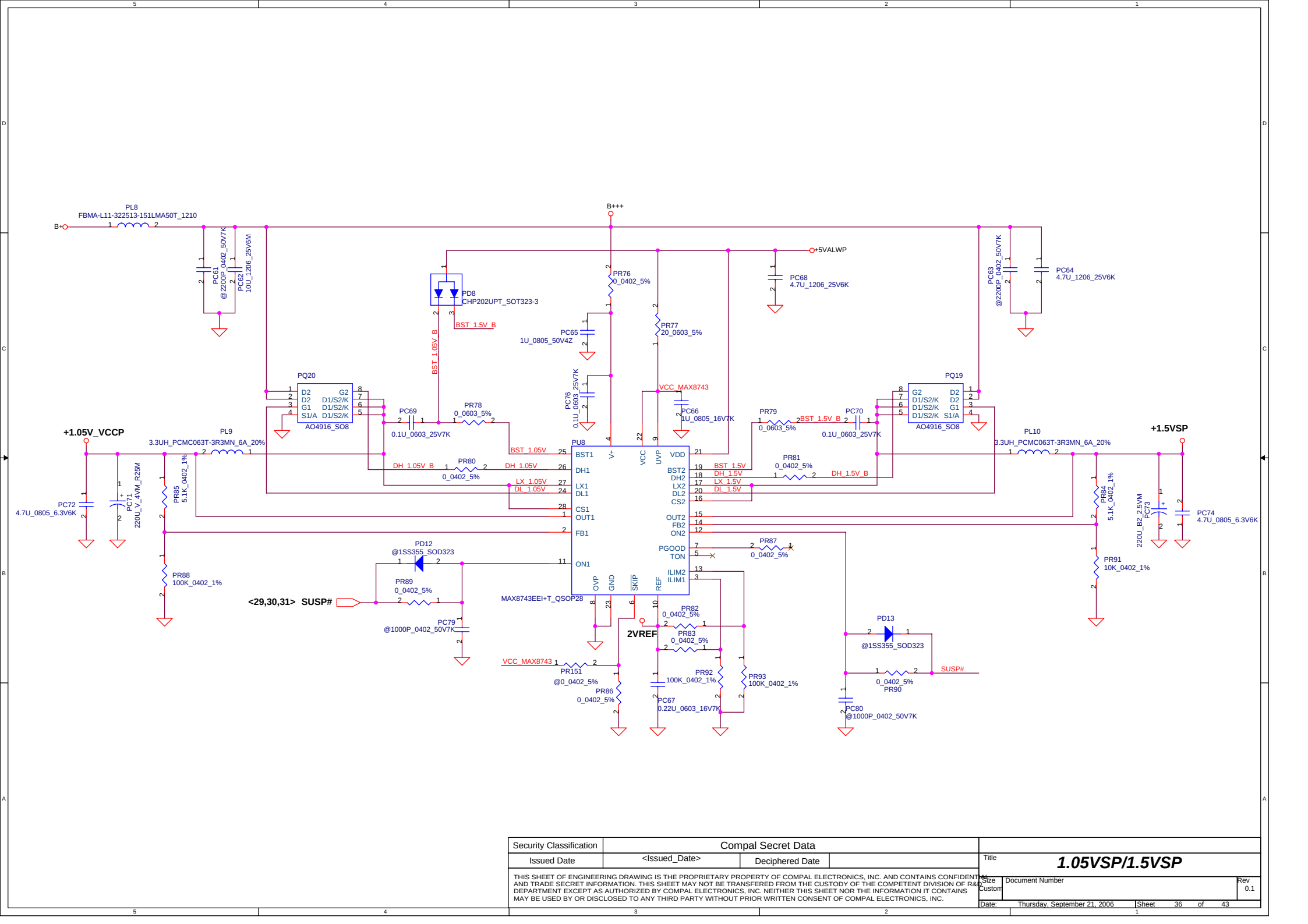




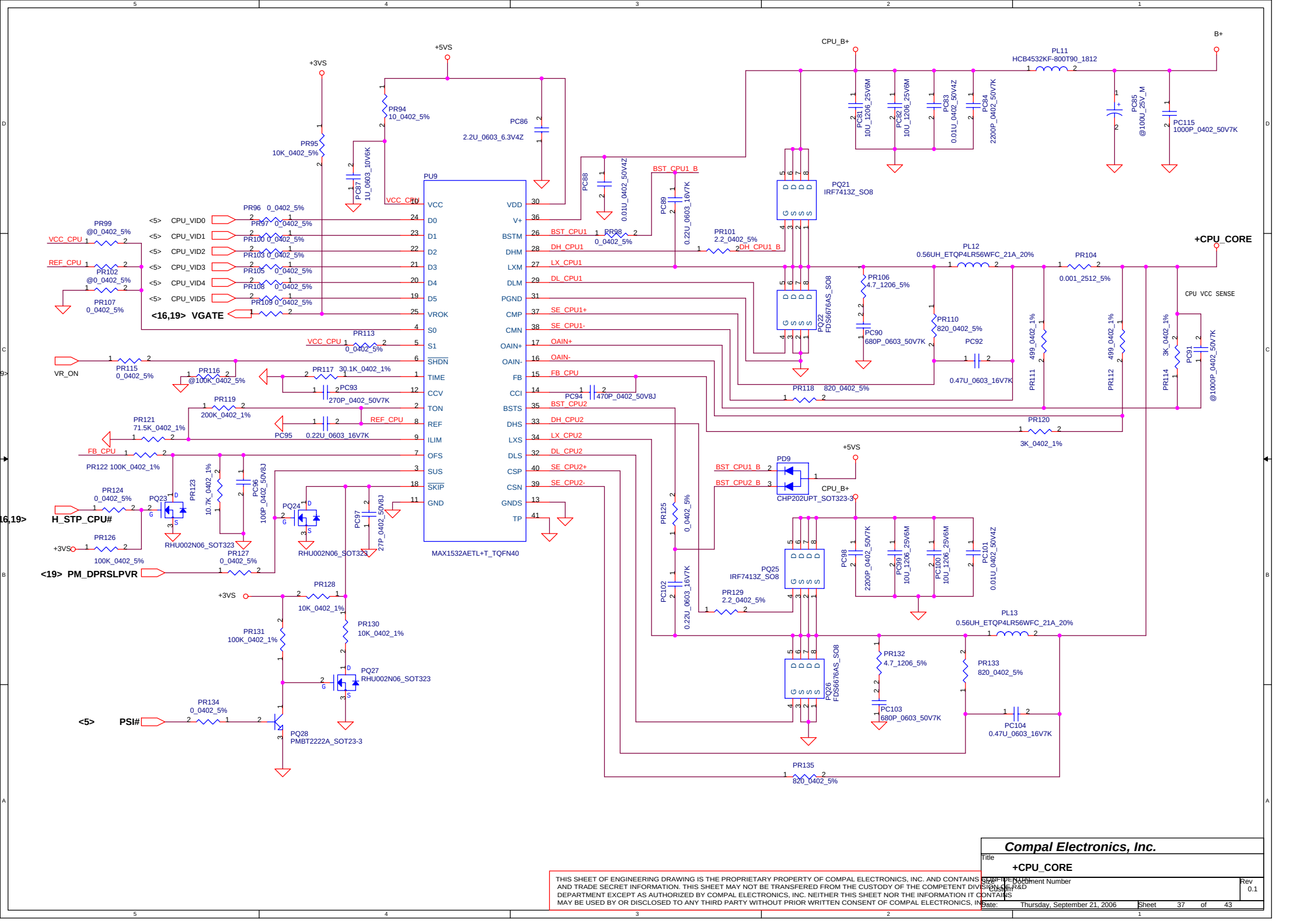
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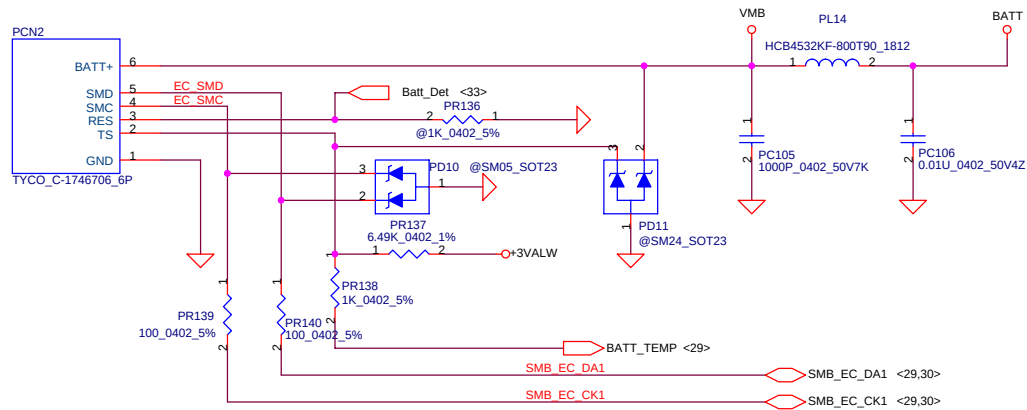


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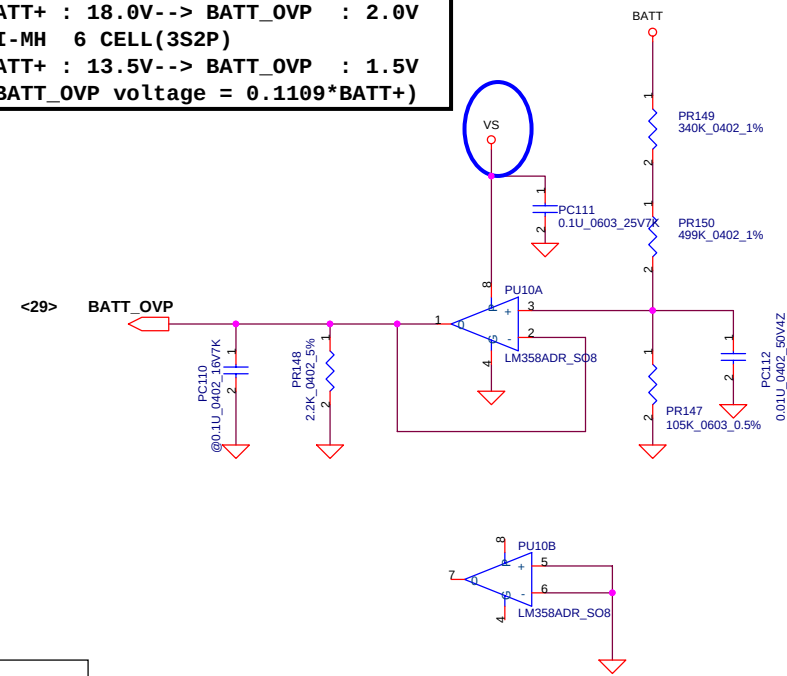


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Title	+CPU_CORE		
Drawing Number	016R60		
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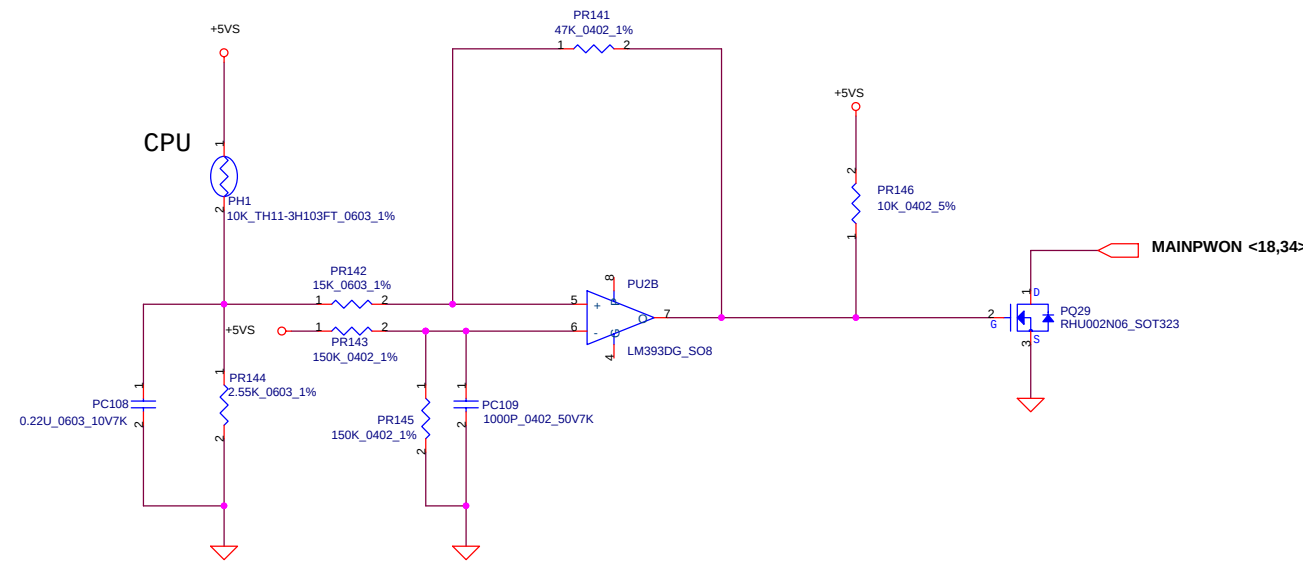
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OVP voltage :
LI-MH 4 CELL(4S1P)/ 8 CELL (4S2P)
BATT+ : 18.0V--> BATT_OVP : 2.0V
LI-MH 6 CELL(3S2P)
BATT+ : 13.5V--> BATT_OVP : 1.5V
(BATT_OVP voltage = 0.1109*BATT+)



PH1 under CPU botten side :
CPU thermal protection at 90 +-3 degree C
Recovery at 47 +-3 degree C



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Version change list (P.I.R. List)

Item	Fixed Issue	Reason for change	Rev.	PG#	Modify List	Note	Phase
1			0.2	6	Change C94,C95,C286 from SGA20331D80 to SGA20331D20	Debug DB	
2		Change U31 to LF parts	0.2	25	Change U31 from SA204680000 to SA204680010	Debug DB	
3		LID switch material error	0.2	30	Change SW3 from SN511000300 to SN111000207	Debug DB	
4		CardBus controller material error	0.2	23	Change U38 from SA014100310 to SA014100130	Debug DB	
5		Un-install some bridge between AGND and DGND	0.2	25	Un-install R1934,R1936,R1938,R1940	Debug DB	
6		Let XM1T# match SW's GPIO definition	0.2	24	Delete Q46,R1913,R1914	Debug DB	
7		Modify Head phone sense's level	0.2	27	Delete R1990 ; Change C1526 to 0.1u 0402 cap ; Change R1991 to 20K resistor.	Debug DB	
8		Reserve for SB internal +1.5VALW regulator	0.2	18	Add R81 and R50	Debug DB	
9		HDD and ODD can't work at same time	0.2	21	Add R27 pull high	Debug DB	
10							
11		DB phase loss	0.2	23	JP54(PCMCIA connector) pin 71, pin72 connect to GND	Debug DB	
12		XM1T# Reserved pull high	0.2	24	Reserve R1913 pull high +3VALW		
13		DB phase error	0.2	17	Change RP43, RP44, RP45 package SIZE from 0804 to1206		
14		Cost down	0.2	19	ChangeR732, R733, R737, R739 to RP46		
15		Cost down	0.2	17	ChangeR277, R274, R265, R279 to RP47		
16		Cost down	0.2	17	ChangeR269, R268, R245, R262 to RP48		
17		Reserver a resistor for 2nd source Amp	0.2	27	Add R28 for 2nd source Amp		
18		Avoid DISPLAYOFF# error status to high	0.2	14	Change R120 to 47K and R9 to 4.7K		
19		Avoid DISPLAYOFF# error status to high	0.2	14	Remove R4, C1 and Add C2. Change pull-up RES R8 with +3VALW to Q1 gate net		
20		EMI Request	0.2	23	Change C641 from 0.1u to 680P, Add C1538(680p) for +3V_CB		
21		EMI Request	0.2	23	Change J4, J6 to bead L37, L38(SM010016810)		
22		EMI Request	0.2	31	Add C1540, C1541, C1542, C1543, C1544 for +1.8V cross plane		
23		EMI Request	0.2	14	Add C1533, C1534, C1535, C1536, C1537 for LVDS EMI Solution		
24		EMI Request	0.2	14	Change L35 from R to Bead		
25		EMI Request	0.2	25	Change R1946, R1945 to Bead L40, L41 and Add L39 For codec regulator		
26		EMI Request	0.2	27	Add C1492, C1493		
27		EMI Request	0.2	27	Change R1982, R1987, R1989 to Bead L42, L43, L44		
28							

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Version change list (P.I.R. List)

Item	Fixed Issue	Reason for change	Rev.	PG#	Modify List	B.Ver#	Phase
1	LAN signal measure fail on Intel site	Delete EMI capacitor	1.0	22	Change C1516,C1517 to NI		
2	WLAN LED work wrong	Add two resister to select WL_LED and XMIT#	1.0	24	Add R2021,R2022 ; R2022 NI ,R2021 0 ohm		
3		increase RTC battery life	1.0	18	D42 pin2 change power form +3VALW to +3VLP		
4	EMITest fail from LAN	Follow intel reference design	1.0	22	C1517 change form 0.01u to 0.1U and C1516 NI		
5	EMITest fail from LAN	Add parallel damping test is PASS but marging	1.0	18、 22			
6	MIC always on have noise	Add MIC_Sense pin to detect MIC plug in	1.0	25	R1954 change from 10k to 0 ohm		
7	MIC always on have noise	Add MIC_Sense pin to detect MIC plug in	1.0	27	Add R ? and C ? , MIC connector pin 3、 pin 5 connect to AGND		
8	M/B ID	Change to 3.3V for MV	1.0	29	R1997 Change to NI		
9	reduce S3 power consumption	Disable wake on LAN function	1.0	19 20 22	Change R16,R11R70 to NI;R15,R12,R71 0 hom		
10	AMON capacitor duplication	C1533,C1552 replace with MC906,MC908	1.0	26	Change MC906,MC908 NI		
11							
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Version change list (P.I.R. List)

Power section

Page 1 of 2

Item	Reason for change	PG#	Modify List	Date	Phase
1	Fine tune the 1.8V OCP set-point as 7.5A (minimum continue load)	35	Change PR68 From SD034182280 (S RES 1/16W 18.2K +-1% 0402) To SD034154280 (S RES 1/16W 15.4K +-1% 0402)	2006/05/26	Before DB
2	Fine tune the charge current as 3A (maximum)	33	Change PR20 From SD034137380 (S RES 1/16W 137K +-1% 0402) To SD034100380 (S RES 1/16W 100K +-1% 0402)	2006/05/30	Before DB
3	Fine tune the AC detector set-point	33	Change PR26 From SD034681380 (S RES 1/16W 681K +-1% 0402) To SD028100480 (S RES 1/16W 1M +-5% 0402)	2006/05/30	Before DB
4	Fine tune the 1.05V OCP set-point as 8.125A (minimum continue load)	36	Change PR82 From SD034274180 (S RES 1/16W 2.74K +-1% 0402) To SD000009480 (S RES 1/16W 1.47K +-1% 0402) Change PR92 From SD034909280 (S RES 1/16W 90.9K +-1% 0402) To SD034499280 (S RES 1/16W 49.9K +-1% 0402)	2006/05/30	Before DB
5	Fine tune the 1.5V OCP set-point as 5A (minimum continue load)	36	Change PR83 From SD034274180 (S RES 1/16W 2.74K +-1% 0402) To SD000009480 (S RES 1/16W 1.47K +-1% 0402) Change PR93 From SD034909280 (S RES 1/16W 90.9K +-1% 0402) To SD034715280 (S RES 1/16W 71.5K +-1% 0402)	2006/05/30	Before DB
6	ID pin for 4 cells battery	38	Add PR136 SD028100180 (S RES 1/16W 1K +-5% 0402)	2006/05/30	Before DB
7	PL1 crack issue	32	As manufactory's comments, change the PL1 from multi-layer bead to beadcore	2006/07/13	DB
8	Modify the sequence of 3V/5V when DC mode	34	Connect the PQ30.1 from VS to +3VLP Change PC41 from 0.1u to 0.047u	2006/07/10	DB
9	The transient of +1.8V is fail	35	Change PR69 from 49.9K to 90.9K for transient	2006/07/12	DB
10	Due to the unstable of ISL6227, change the +1.5V/ +1.05V solution to MAX8743	36	Change the +1.5V/ +1.05V solution from ISL6227 to MAX8743, included all the related components.	2006/07/7	DB
11	For EMI's concern	35	Add PC116 (680pf)	2006/07/17	DB
12	For EMI's concern	37	PC115 (1000pf), PR106/PR132 (4.7ohm), and PC90/ PC103 (680pf)	2006/07/17	DB
13	Modify the sequence of 1.8V for S3 can't resume issue	35	Change connection of PR64.1 from +5VS to +5VALW	2006/08/02	SI
14	For 4 series/ 3 series battery selection, add circuit for changing charge voltage as 16.8V (4 series) or 12.6V (3 series)	33	Add PQ33, PQ34, PR154, PR155 and PR156	2006/08/10	SI
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Version change list (P.I.R. List)

Power section

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Item	Reason for change	PG#	Modify List	Date	Phase
15	Disable the "ALARM" function of charger	33	Un-pop PR35, PR39 and PC23 Change PR28 from 604K to 100K Change connection of PR28.1 from BATT to +3VLP	2006/08/16	SI
16	Reduce S3 power consumption	38	Change the connection of PU10.1 from VS to VL	2006/08/16	SI
17	For EMI's requirement	35	Add PR152 (4.7ohm) and PC114 (680pf)	2006/08/16	SI
18	Item 16 causes the S4 power consumption fail, so return to SI design. This change will not impact the S3 power consumption (this change adds less than 10mW back)	38	Change the connection of PU10.1 from VL to VS	2006/09/05	PV
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